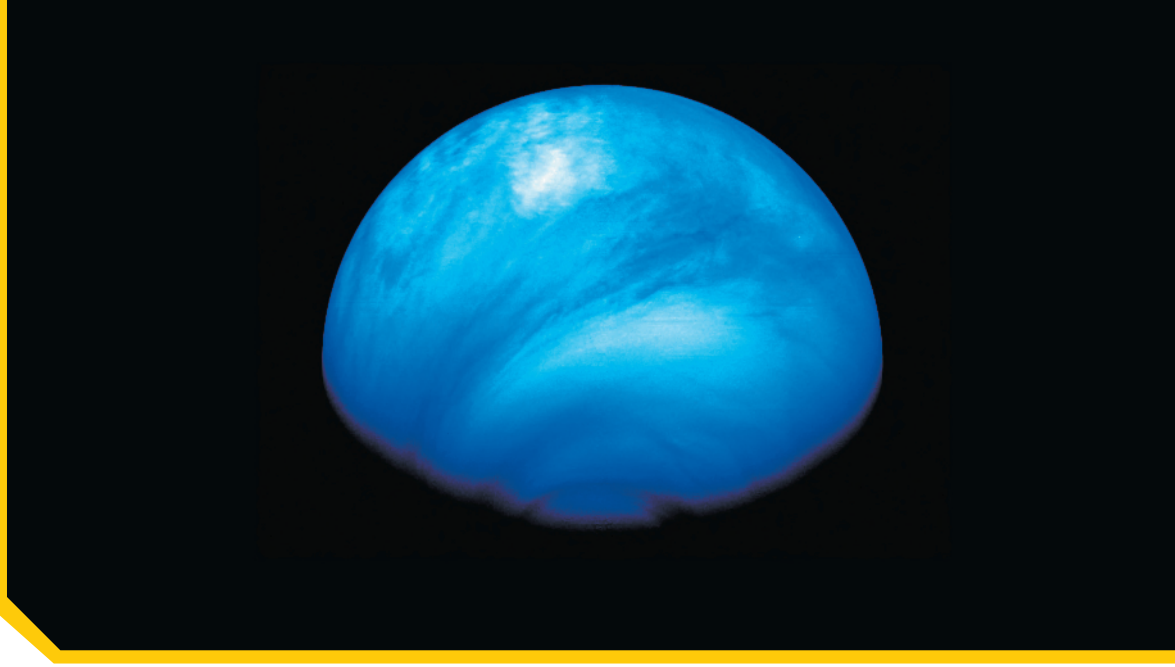
? Although Venus is almost twice as far as Mercury is from the sun, it has a higher surface temperature: 735 K (462°C = 863°F). The reason is that Venus has a thick, cloud-shrouded atmosphere (shown here in false color) that is 96.5% carbon dioxide (CO_2). Molecules of CO_2 are a potent agent for raising Venus's temperature because (i) they absorb infrared radiation in vibrational transitions; (ii) they absorb infrared radiation in electronic transitions; (iii) they absorb ultraviolet radiation in vibrational transitions; (iv) they absorb ultraviolet radiation in electronic transitions; (v) more than one of these.



42 Molecules and Condensed Matter

LEARNING OUTCOMES

In this chapter, you'll learn...

42.1 The various types of bonds that hold atoms together.
42.2 How molecular spectra reveal the rotational and vibrational dynamics of molecules.
42.3 How and why atoms form into crystalline structures.
42.4 How to use the energy-band concept to explain the electrical properties of solids.
42.5 A model that explains many of the physical properties of metals.
42.6 How a small amount of an impurity can radically affect the character of a semiconductor.
42.7 Some of the technological applications of semiconductor devices.
42.8 Why certain materials become superconductors at low temperature.

You'll need to review...

10.5 Rotational kinetic energy.
17.6 Greenhouse effect.
18.3–18.5 Ideal gases; equipartition of energy; Maxwell–Boltzmann distribution.
23.1 Electric potential energy.
24.4, 24.5 Dielectrics.
25.2 Resistivity.
39.3 Reduced mass.
40.5 Quantum-mechanical harmonic oscillators.
41.2–41.4, 41.6 Particle in a 3-D box; rotational energy levels; selection rules; exclusion principle.

Isolated atoms, which we studied in Chapter 41, are the exception; usually we find atoms combined to form molecules or more extended structures we call condensed matter (liquid or solid). In this chapter we'll study the attractive forces, called molecular bonds, that cause atoms to combine into molecules. We'll see that just as atoms have quantized energies determined by the quantum-mechanical state of their electrons, so molecules have quantized energies determined by their rotational and vibrational states.

The same physical principles behind molecular bonds also apply to the study of condensed matter, in which various types of bonding occur. We'll explore the concept of energy bands and see how it helps us understand the properties of solids. Then we'll look more closely at the properties of a special class of solids called semiconductors. Devices using semiconductors are found in every mobile phone, TV, and computer used today.

42.1 TYPES OF MOLECULAR BONDS

We can use our discussion of atomic structure in Chapter 41 as a basis for exploring the nature of *molecular bonds,* the interactions that hold atoms together to form stable structures such as molecules and solids.

Ionic Bonds

The **ionic bond** is an interaction between oppositely charged *ionized* atoms. The most familiar example is sodium chloride (NaCl), in which the sodium (Na) atom loses its one 3*s* electron, which fills the vacancy in the 3*p* subshell of the chlorine (Cl) atom.

Let's look at the energy balance in this transaction. Removing the 3*s* electron from a neutral Na atom requires 5.138 eV of energy; this is called the *ionization energy* of Na. The neutral Cl atom can attract an extra electron into the vacancy in the 3*p* subshell, where it is incompletely screened by the other electrons and therefore is attracted

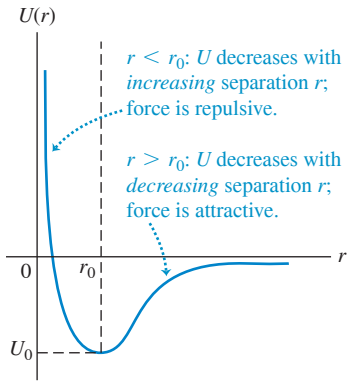
to the nucleus. This state has 3.613 eV lower energy than a state with a neutral Cl atom and a distant free electron; 3.613 eV is the magnitude of the *electron affinity* of chlorine. Thus creating an Na^+ ion and a Cl^- ion that are well separated from each other (so they do not interact) requires a net investment of only 5.138 eV − 3.613 eV = 1.525 eV.

When their mutual attraction brings the Na^+ and Cl^- ions together, the magnitude of their negative potential energy is determined by their separation r (**Fig. 42.1**). The exclusion principle (Section 41.6), which states that only one electron can occupy a given quantum-mechanical state, limits how small this separation can be. As r decreases, the exclusion principle distorts the charge clouds, so the ions no longer interact like point charges and the interaction eventually becomes repulsive.

The minimum electric potential energy for NaCl turns out to be −5.7 eV at a separation of 0.24 nm. The net energy released in creating the ions and letting them come together to the equilibrium separation of 0.24 nm is 5.7 eV − 1.525 eV = 4.2 eV. Thus, if we ignore the kinetic energy of the ions, 4.2 eV is the *binding energy* of the NaCl molecule, the energy that is needed to dissociate the molecule into separate neutral atoms.

Ionic bonds can involve more than one electron per atom. For instance, alkaline earth elements form ionic compounds in which an atom loses *two* electrons; an example is magnesium chloride, or $Mg^{2+}(Cl^-)_2$. Ionic bonds that involve a loss of more than two electrons are relatively rare. Instead, a different kind of bond, the *covalent* bond, comes into operation. We'll discuss this type of bond below.

Figure **42.1** When the separation r between two oppositely charged ions is large, the potential energy $U(r)$ is proportional to $1/r$ as for point charges and the force is attractive. As r decreases, the charge clouds of the two atoms overlap and the force becomes less attractive. If r is less than the equilibrium separation r_0, the force is repulsive.



EXAMPLE 42.1 Electric potential energy of the NaCl molecule

Find the electric potential energy of an Na^+ ion and a Cl^- ion separated by 0.24 nm. Consider the ions as point charges.

IDENTIFY and SET UP Equation (23.9) in Section 23.1 tells us that the electric potential energy of two point charges q and q_0 separated by a distance r is $U = qq_0/4\pi\epsilon_0 r$.

EXECUTE We have $q = +e$ (for Na^+), $q_0 = -e$ (for Cl^-), and $r = 0.24\text{ nm} = 0.24 \times 10^{-9}\text{ m}$. From Eq. (23.9),

$$U = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r_0} = -(9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)\frac{(1.6 \times 10^{-19}\ \text{C})^2}{0.24 \times 10^{-9}\ \text{m}} = -9.6 \times 10^{-19}\ \text{J} = -6.0\ \text{eV}$$

EVALUATE This result agrees fairly well with the observed value of −5.7 eV. The reason for the difference is that when the two ions are at their equilibrium separation of 0.24 nm, the outer regions of their electron clouds overlap. Hence the two ions don't behave exactly like point charges.

KEYCONCEPT In an ionic bond, the electric potential energy of the two ions (one positive and one negative) is negative and roughly the same as if we consider the two ions as point charges.

Covalent Bonds

Unlike the transaction that occurs in an ionic bond, in a **covalent bond** there is no net transfer of electrons from one atom to another. The simplest covalent bond is found in the hydrogen molecule, a structure containing two protons and two electrons. As the separate atoms (**Fig. 42.2a**) come together, the electron wave functions are distorted and become more concentrated in the region between the two protons (Fig. 42.2b). The net attraction of the electrons for each proton more than balances the repulsion of the two protons and of the two electrons.

The attractive interaction is then supplied by a *pair* of electrons, one contributed by each atom, with charge clouds that are concentrated primarily in the region between the two atoms. The energy of the covalent bond in the hydrogen molecule H_2 is −4.48 eV.

As we saw in Section 41.6, the exclusion principle permits two electrons to occupy the same region of space (that is, to be in the same spatial quantum state) only when they have opposite spins. Hence the two electrons in the H_2 covalent bond (Fig. 42.2b) must have opposite spins, since both occupy the same region between the two nuclei. Opposite spins are an essential requirement for a covalent bond, and no more than two electrons can participate in such a bond.

Figure **42.2** Covalent bond in a hydrogen molecule.

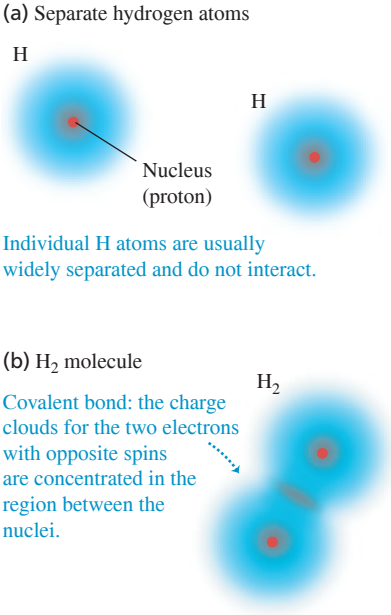
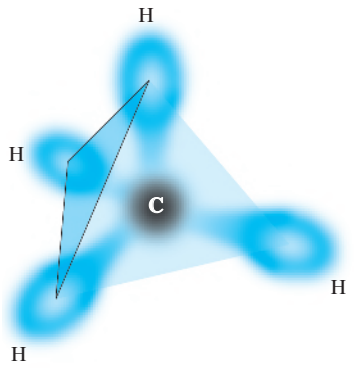


Figure 42.3 Schematic diagram of the methane (CH_4) molecule. The carbon atom is at the center of a regular tetrahedron and forms four covalent bonds with the hydrogen atoms at the corners. Each covalent bond includes two electrons with opposite spins, forming a charge cloud that is concentrated between the carbon atom and a hydrogen atom.



However, an atom with several electrons in its outermost shell can form several covalent bonds. The bonding of carbon and hydrogen atoms, of central importance in organic chemistry, is an example. In the *methane* molecule (CH_4) the carbon atom is at the center of a regular tetrahedron, with a hydrogen atom at each corner. The carbon atom has four electrons in its L shell, and each of these four electrons forms a covalent bond with one of the four hydrogen atoms (**Fig. 42.3**). Similar patterns occur in more complex organic molecules.

Covalent bonds are highly directional. In the methane molecule the wave function for each of carbon's four valence electrons is a combination of the $2s$ and $2p$ wave functions called a *hybrid wave function*. The probability distribution for each one has a lobe protruding toward a corner of a tetrahedron. This symmetric arrangement minimizes the overlap of wave functions for the electron pairs, which in turn minimizes the positive potential energy associated with repulsion between the pairs.

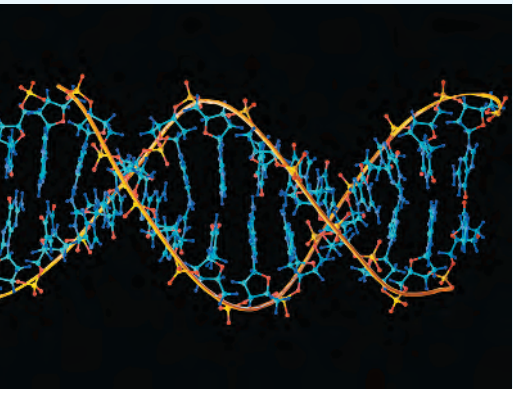
Ionic and covalent bonds represent two extremes in molecular bonding, but there is no sharp division between the two types. Often there is a *partial* transfer of one or more electrons from one atom to another. As a result, many molecules that have dissimilar atoms have electric dipole moments—that is, a preponderance of positive charge at one end and of negative charge at the other. Such molecules are called *polar* molecules. Water molecules have large electric dipole moments; these are responsible for the exceptionally large dielectric constant of liquid water (see Sections 24.4 and 24.5).

Van der Waals Bonds

Ionic and covalent bonds, with typical bond energies of 1 to 5 eV, are called *strong bonds*. There are also two types of weaker bonds. One of these, the **van der Waals bond,** is an interaction between the electric dipole moments of atoms or molecules; typical energies are 0.1 eV or less. The bonding of water molecules in the liquid and solid states results partly from dipole–dipole interactions.

No atom has a permanent electric dipole moment, nor do many molecules. However, fluctuating charge distributions can lead to fluctuating dipole moments; these in turn can induce dipole moments in neighboring structures. Overall, the resulting dipole–dipole interaction is attractive, giving a weak bonding of atoms or molecules. The interaction potential energy drops off very quickly with distance r between molecules, usually as $1/r^6$. The liquefaction and solidification of the inert gases and of molecules such as H_2, O_2, and N_2 are due to induced-dipole van der Waals interactions. Not much thermal-agitation energy is needed to break these weak bonds, so such substances usually exist in the liquid and solid states only at very low temperatures.

BIO APPLICATION Molecular Zipper
A DNA molecule functions like a twisted zipper. Each of the two strands of the "zipper" consists of an outer backbone and inward-facing nucleotide "teeth"; hydrogen bonds between facing teeth "zip" the strands together. The covalent bonds that hold together the atoms of each strand are strong, whereas the hydrogen bonds are relatively weak, so that the cell's biochemical machinery can easily separate the strands for reading or copying.



Hydrogen Bonds

In the other type of weak bond, the **hydrogen bond,** a proton (H^+ ion) gets between two atoms, polarizing them and attracting them by means of the induced dipoles. This bond is unique to hydrogen-containing compounds because only hydrogen has a singly ionized state with no remaining electron cloud; the hydrogen ion is a bare proton, much smaller than any other singly ionized atom. The bond energy is usually less than 0.5 eV. The hydrogen bond is responsible for the cross-linking of long-chain organic molecules such as polyethylene (used in plastic bags). Hydrogen bonding also plays a role in the structure of ice.

All these bond types hold the atoms together in *solids* as well as in molecules. Indeed, a solid is in many respects a giant molecule. Still another type of bonding, the *metallic bond,* comes into play in the structure of metallic solids. We'll return to this subject in Section 42.3.

TEST YOUR UNDERSTANDING OF SECTION 42.1 If electrons obeyed the exclusion principle but did *not* have spin, how many electrons could participate in a covalent bond? (i) One; (ii) two; (iii) three; (iv) more than three.

ANSWER (i) The exclusion principle states that only one electron can be in a given state. Real electrons have spin, so two electrons (one spin up, one spin down) can be in a given *spatial* state and hence two can participate in a given covalent bond between two atoms. If electrons obeyed the exclusion principle but did not have spin, that state of an electron would be completely described by its spatial distribution and only *one* electron could participate in a covalent bond. (We'll learn in Chapter 44 that this situation is wholly imaginary: There are subatomic particles without spin, but they do *not* obey the exclusion principle.)

42.2 MOLECULAR SPECTRA

Molecules have energy levels that are associated with rotation of a molecule as a whole and with vibration of the atoms relative to each other. Just as transitions between energy levels in atoms lead to atomic spectra, transitions between rotational and vibrational levels in molecules lead to *molecular spectra*.

Rotational Energy Levels

In this discussion we'll concentrate mostly on *diatomic* molecules, to keep things as simple as possible. In **Fig. 42.4** we picture a diatomic molecule as a rigid dumbbell (two point masses m_1 and m_2 separated by a constant distance r_0) that can *rotate* about axes through its center of mass, perpendicular to the line joining them. What are the energy levels associated with this motion?

We showed in Section 10.5 that when a rigid body rotates with angular speed ω about a perpendicular axis through its center of mass, the magnitude L of its angular momentum is given by Eq. (10.28), $L = I\omega$, where I is its moment of inertia for that axis. Its kinetic energy is given by Eq. (9.17), $K = \frac{1}{2}I\omega^2$. Combining these two equations, we find $K = L^2/2I$. There is no potential energy U, so the kinetic energy K is equal to the total mechanical energy E:

$$E = \frac{L^2}{2I} \tag{42.1}$$

Zero potential energy means that U does not depend on the angular coordinate of the molecule. But the potential-energy function U for the hydrogen atom (see Section 41.3) also has no dependence on angular coordinates. Thus the angular solutions to the Schrödinger equation for rigid-body rotation are the same as for the hydrogen atom, and the angular momentum is quantized in the same way. As in Eq. (41.22),

$$L = \sqrt{l(l+1)}\,\hbar \qquad (l = 0,\ 1,\ 2, \ldots) \tag{42.2}$$

Combining Eqs. (42.1) and (42.2), we obtain the *rotational energy levels:*

Rotational energy levels of a diatomic molecule

$$E_l = l(l+1)\frac{\hbar^2}{2I} \tag{42.3}$$

(l: Rotational quantum number ($l = 0, 1, 2, \ldots$); $\hbar^2$: Planck's constant divided by 2π; I: Moment of inertia for axis through molecule's cm)

Figure 42.5 is an energy-level diagram showing these rotational levels. The $l = 0$ ground level has zero angular momentum (no rotation and zero rotational energy E). The spacing of adjacent levels increases with increasing l.

We can express the moment of inertia I in Eqs. (42.1) and (42.3) in terms of the *reduced mass* m_r of the molecule:

Figure **42.4** A diatomic molecule modeled as two point masses m_1 and m_2 separated by a distance r_0. The distances of the masses from the center of mass are r_1 and r_2, where $r_1 + r_2 = r_0$.

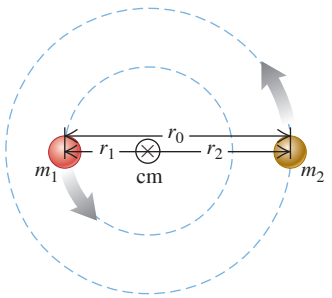
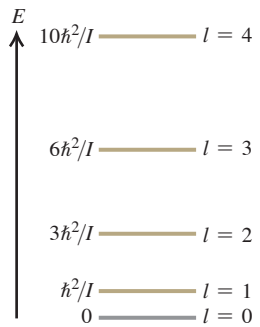


Figure **42.5** The ground level and first four excited rotational energy levels for a diatomic molecule. The levels are not equally spaced.



Reduced mass of a diatomic molecule

$$m_r = \frac{m_1 m_2}{m_1 + m_2} \qquad (42.4)$$

(m_1: mass of atom 1; m_2: mass of atom 2)

We introduced the reduced mass in Section 39.3 to accommodate the finite nuclear mass of the hydrogen atom. In Fig. 42.4 the distances r_1 and r_2 are the distances from the center of mass to the centers of the atoms. By the definition of the center of mass, $m_1 r_1 = m_2 r_2$, and the figure also shows that $r_0 = r_1 + r_2$. Solving these equations for r_1 and r_2, we find

$$r_1 = \frac{m_2}{m_1 + m_2} r_0 \qquad r_2 = \frac{m_1}{m_1 + m_2} r_0 \qquad (42.5)$$

The moment of inertia is $I = m_1 r_1^2 + m_2 r_2^2$; substituting Eq. (42.5), we find

$$I = m_1 \frac{m_2^2}{(m_1 + m_2)^2} r_0^2 + m_2 \frac{m_1^2}{(m_1 + m_2)^2} r_0^2 = \frac{m_1 m_2}{m_1 + m_2} r_0^2 \quad \text{or}$$

Moment of inertia of a diatomic molecule, axis through molecule's cm

$$I = m_r r_0^2 \qquad (42.6)$$

(m_r: reduced mass; r_0: distance between centers of molecule's two atoms)

The moment of inertia is the same as that of an equivalent *single* point mass m_r that orbits the axis in a circle of radius r_0.

To conserve angular momentum and account for the angular momentum of the emitted or absorbed photon, the allowed transitions between rotational states must satisfy the same selection rule that we discussed in Section 41.4 for allowed transitions between the states of an atom: l must change by exactly one unit; that is, $\Delta l = \pm 1$.

EXAMPLE 42.2 Rotational spectrum of carbon monoxide

WITH √ARIATION PROBLEMS

The two nuclei in the carbon monoxide (CO) molecule are 0.1128 nm apart. The mass of the most common carbon atom is 1.993×10^{-26} kg; that of the most common oxygen atom is 2.656×10^{-26} kg. (a) Find the energies of the lowest three rotational energy levels of CO. Express your results in meV ($1\ \text{meV} = 10^{-3}\ \text{eV}$). (b) Find the wavelength of the photon emitted in the transition from the $l = 2$ to the $l = 1$ level.

IDENTIFY and SET UP This problem uses the ideas developed in this section about the rotational energy levels of molecules. We are given the distance r_0 between the atoms and their masses m_1 and m_2. We find the reduced mass m_r from Eq. (42.4), the moment of inertia I from Eq. (42.6), and the energies E_l from Eq. (42.3). The energy E of the emitted photon is equal to the difference in energy between the $l = 2$ and $l = 1$ levels. (This transition obeys the $\Delta l = \pm 1$ selection rule, since $\Delta l = 1 - 2 = -1$.) We determine the photon wavelength by using $E = hc/\lambda$.

EXECUTE (a) From Eqs. (42.4) and (42.6), the reduced mass and moment of inertia of the CO molecule are:

$$m_r = \frac{m_1 m_2}{m_1 + m_2}$$

$$= \frac{(1.993 \times 10^{-26}\ \text{kg})(2.656 \times 10^{-26}\ \text{kg})}{(1.993 \times 10^{-26}\ \text{kg}) + (2.656 \times 10^{-26}\ \text{kg})} = 1.139 \times 10^{-26}\ \text{kg}$$

$$I = m_r r_0^2$$

$$= (1.139 \times 10^{-26}\ \text{kg})(0.1128 \times 10^{-9}\ \text{m})^2 = 1.449 \times 10^{-46}\ \text{kg} \cdot \text{m}^2$$

The rotational levels are given by Eq. (42.3):

$$E_l = l(l+1)\frac{\hbar^2}{2I} = l(l+1)\frac{(1.0546 \times 10^{-34}\ \text{J} \cdot \text{s})^2}{2(1.449 \times 10^{-46}\ \text{kg} \cdot \text{m}^2)}$$

$$= l(l+1)(3.838 \times 10^{-23}\ \text{J}) = l(l+1)0.2395\ \text{meV}$$

($1\ \text{meV} = 10^{-3}\ \text{eV}$.) Substituting $l = 0, 1, 2$, we find

$$E_0 = 0 \qquad E_1 = 0.479\ \text{meV} \qquad E_2 = 1.437\ \text{meV}$$

(b) The photon energy and wavelength are

$$E = E_2 - E_1 = 0.958\ \text{meV}$$

$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\ \text{eV} \cdot \text{s})(3.00 \times 10^8\ \text{m/s})}{0.958 \times 10^{-3}\ \text{eV}}$$

$$= 1.29 \times 10^{-3}\ \text{m} = 1.29\ \text{mm}$$

EVALUATE The differences between the first few rotational energy levels of CO are very small (about $1\ \text{meV} = 10^{-3}\ \text{eV}$) compared to the differences between atomic energy levels (typically a few eV). Hence a photon emitted by a CO molecule in a transition from the $l = 2$ to the $l = 1$ level has very low energy and a very long wavelength compared to the visible light emitted by excited atoms. Photon wavelengths for rotational transitions in molecules are typically in the microwave and far infrared regions of the spectrum.

In this example we were given the equilibrium separation between the atoms, also called the *bond length,* and we used it to calculate one of the wavelengths emitted by excited CO molecules. In experiments, scientists work this problem backward: By measuring the long-wavelength emissions of a sample of diatomic molecules, they determine the moment of inertia of the molecule and hence the bond length.

KEYCONCEPT Because angular momentum is quantized (quantum number l), the rotational kinetic energy of a diatomic molecule is quantized as well. This energy is proportional to $l(l+1)$ and inversely proportional to the moment of inertia of the molecule for an axis through the molecule's center of mass. Transitions between rotational states that involve photon emission or absorption are allowed only if l changes by 1.

Vibrational Energy Levels

Molecules are never completely rigid. In a more realistic model of a diatomic molecule we represent the connection between atoms not as a rigid rod but as a spring (**Fig. 42.6**). Then, in addition to rotating, the atoms of the molecule can *vibrate* about their equilibrium positions along the line joining them. For small oscillations the restoring force can be taken as proportional to the displacement from the equilibrium separation r_0 (like a spring that obeys Hooke's law with a force constant k'), and the system is a harmonic oscillator. We discussed the quantum-mechanical harmonic oscillator in Section 40.5. The energy levels are given by Eq. (40.46) with the mass m replaced by the reduced mass m_r:

Vibrational energy levels of a diatomic molecule

$$E_n = (n + \tfrac{1}{2})\hbar\omega = (n + \tfrac{1}{2})\hbar\sqrt{\frac{k'}{m_r}} \qquad (42.7)$$

Vibrational quantum number ($n = 0, 1, 2, \ldots$)

Planck's constant divided by 2π

Oscillation angular frequency

Reduced mass

Force constant

The spacing in energy between any two adjacent vibrational levels is

$$\Delta E = \hbar\omega = \hbar\sqrt{\frac{k'}{m_r}} \qquad (42.8)$$

Figure 42.7 is an energy-level diagram showing these vibrational levels. As an example, for the CO molecule of Example 42.2 the spacing $\hbar\omega$ between levels is 0.2690 eV. From Eq. (42.8) this corresponds to a force constant of 1.90×10^3 N/m, which is a fairly loose spring. (To stretch a macroscopic spring with this value of k' by 1.0 cm would require a force of only 19 N.) Force constants for diatomic molecules are typically about 100 to 2000 N/m.

CAUTION **Watch out for k, k', and K** As in Section 40.5 we're using k' for the force constant, this time to minimize confusion with Boltzmann's constant k, the gas constant per molecule (introduced in Section 18.3). Besides the quantities k and k', we also use the absolute temperature unit $1\,\text{K} = 1$ kelvin.

Rotation and Vibration Combined

Visible-light photons have energies between 1.65 eV and 3.26 eV. The 0.2690 eV energy difference between vibrational levels for carbon monoxide (CO) corresponds to a photon of wavelength 4.613 μm, in the infrared region of the spectrum. This is much closer to the visible region than is the photon in the rotational transition in Example 42.2. In general the energy differences for molecular *vibration* are much smaller than those that produce atomic spectra, but much larger than the energy differences for molecular *rotation*.

When we include *both* rotational and vibrational energies, the energy levels for our diatomic molecule are

$$E_{nl} = l(l+1)\frac{\hbar^2}{2I} + (n + \tfrac{1}{2})\hbar\sqrt{\frac{k'}{m_r}} \qquad (42.9)$$

Figure 42.8 shows the energy-level diagram. For each value of n there are many values of l, forming a series of closely spaced levels.

The red arrows in Fig. 42.8 show several possible transitions in which a molecule goes from a level with $n = 2$ to a level with $n = 1$ by emitting a photon. As we mentioned, these transitions must obey the selection rule $\Delta l = \pm 1$ to conserve angular momentum. Another selection rule states that if the vibrational level changes, the vibrational quantum number n in Eq. (42.9) must increase by 1 ($\Delta n = 1$) if a photon is absorbed or decrease by 1 ($\Delta n = -1$) if a photon is emitted.

Illustrating these selection rules, Fig. 42.8 shows that a molecule in the $n = 2$, $l = 4$ level can emit a photon and drop into the $n = 1$, $l = 5$ level ($\Delta n = -1$, $\Delta l = +1$) or the $n = 1$, $l = 3$ level ($\Delta n = -1$, $\Delta l = -1$), but is forbidden from making a $\Delta n = -1$, $\Delta l = 0$ transition into the $n = 1$, $l = 4$ level.

Figure **42.6** A diatomic molecule modeled as two point masses m_1 and m_2 connected by a spring with force constant k'.

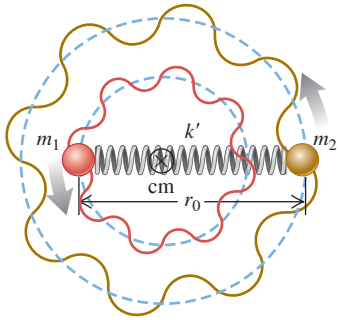


Figure **42.7** The ground level and first three excited vibrational levels for a diatomic molecule, assuming small displacements from equilibrium so we can treat the oscillations as simple harmonic. (Compare Fig. 40.25.)

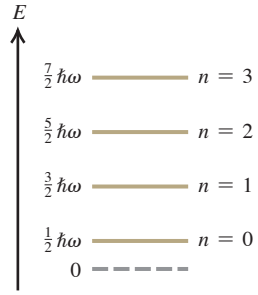


Figure **42.8** Energy-level diagram for vibrational and rotational energy levels of a diatomic molecule. For each vibrational level (n) there is a series of more closely spaced rotational levels (l). Several transitions corresponding to a single band in a band spectrum are shown. These transitions obey the selection rule $\Delta l = \pm 1$.

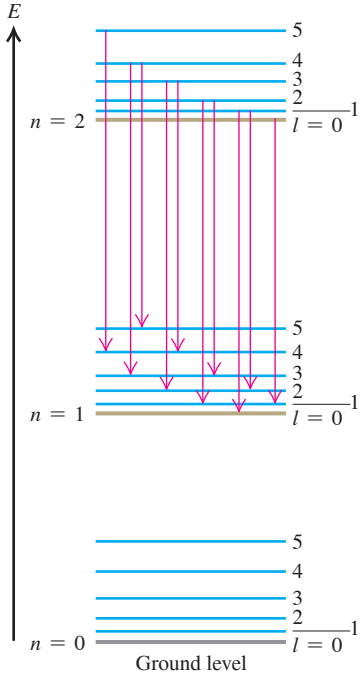
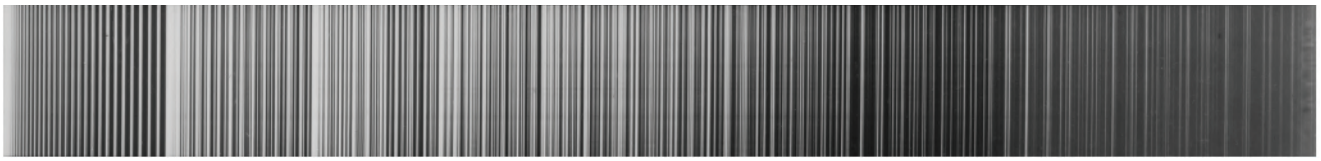


Figure 42.9 A typical molecular band spectrum.



Transitions between states with various pairs of n-values give different series of spectrum lines, and the resulting spectrum has a series of *bands*. Each band corresponds to a particular vibrational transition, and each individual line in a band represents a particular rotational transition, with the selection rule $\Delta l = \pm 1$. **Figure 42.9** shows a typical *band spectrum*.

All molecules can have excited states of the *electrons* in addition to the rotational and vibrational states that we have described. In general, these lie at higher energies than the rotational and vibrational states, and there is no simple rule relating them. When there is a transition between electronic states, the $\Delta n = \pm 1$ selection rule for the vibrational levels no longer holds.

EXAMPLE 42.3 Vibration-rotation spectrum of carbon monoxide

WITH √ARIATION PROBLEMS

Consider again the CO molecule of Example 42.2. Find the wavelength of the photon emitted by a CO molecule when its vibrational energy changes and its rotational energy is (a) initially zero and (b) finally zero.

IDENTIFY and SET UP We need to use the selection rules for the vibrational and rotational transitions of a diatomic molecule. Since a photon is emitted as the vibrational energy changes, the selection rule $\Delta n = -1$ tells us that the vibrational quantum number n decreases by 1 in both parts (a) and (b). In part (a) the initial value of l is zero; the selection rule $\Delta l = \pm 1$ tells us that the *final* value of l is 1, so the rotational energy increases in this case. In part (b) the *final* value of l is zero; $\Delta l = \pm 1$ then tells us that the *initial* value of l is 1, and the rotational energy decreases.

The energy E of the emitted photon is the difference between the initial and final energies of the molecule, accounting for the change in both vibrational and rotational energies. In part (a) E equals the difference $\hbar\omega$ between adjacent vibrational energy levels *minus* the rotational energy that the molecule *gains;* in part (b) E equals $\hbar\omega$ *plus* the rotational energy that the molecule *loses*. Example 42.2 tells us that the difference between the $l = 0$ and $l = 1$ rotational energy levels is 0.479 meV = 0.000479 eV, and we learned above that the vibrational energy-level separation for CO is $\hbar\omega = 0.2690$ eV. We use $E = hc/\lambda$ to determine the corresponding wavelengths (our target variables).

EXECUTE (a) The CO molecule loses $\hbar\omega = 0.2690$ eV of vibrational energy and gains 0.000479 eV of rotational energy. Hence the energy E that goes into the emitted photon equals 0.2690 eV *less* 0.000479 eV, or 0.2685 eV. The photon wavelength is

$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(2.998 \times 10^{8}\ \text{m/s})}{0.2685\ \text{eV}}$$

$$= 4.618 \times 10^{-6}\ \text{m} = 4.618\ \mu\text{m}$$

(b) Now the CO molecule loses $\hbar\omega = 0.2690$ eV of vibrational energy and also loses 0.000479 eV of rotational energy, so the energy that goes into the photon is $E = 0.2690\ \text{eV} + 0.000479\ \text{eV} = 0.2695$ eV. The wavelength is

$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(2.998 \times 10^{8}\ \text{m/s})}{0.2695\ \text{eV}}$$

$$= 4.601 \times 10^{-6}\ \text{m} = 4.601\ \mu\text{m}$$

EVALUATE In part (b) the molecule loses more energy than it does in part (a), so the emitted photon must have greater energy and a shorter wavelength. That's just what our results show.

KEYCONCEPT In addition to rotational motion around its center of mass (quantum number l), a diatomic molecule can have vibrational motion of its atoms. The vibrational energy levels (quantum number n) are evenly spaced in energy. Transitions between vibrational states that involve photon emission or absorption are allowed only if n and l both change by 1.

Complex Molecules

We can apply these same principles to more complex molecules. A molecule with three or more atoms has several different kinds or *modes* of vibratory motion. Each mode has its own set of energy levels, related to its frequency by Eq. (42.7). In nearly all cases the associated radiation lies in the infrared region of the electromagnetic spectrum.

Infrared spectroscopy has proved to be an extremely valuable analytical tool. It provides information about the strength, rigidity, and length of molecular bonds and the structure of complex molecules. Also, because every molecule (like every atom) has its characteristic spectrum, infrared spectroscopy can be used to identify unknown compounds.

One molecule that can readily absorb and emit infrared radiation is carbon dioxide (CO_2). **Figure 42.10** shows the three possible modes of vibration of a CO_2 molecule. A number of transitions are possible between excited levels of the same vibrational mode as well as between levels of different vibrational modes. The energy differences are less than 1 eV in all of these transitions, and so involve infrared photons of wavelength longer than 1 μm. Hence a gas of CO_2 can readily absorb light at a number of different infrared wavelengths. This makes CO_2 primarily responsible for the greenhouse effect (Section 17.6) on the earth, even though CO_2 is only 0.04% of our atmosphere by volume. On Venus, however, the atmosphere has more than 90 times the total mass of our atmosphere and is almost entirely CO_2. The resulting greenhouse effect is tremendous: The surface temperature on Venus is more than 400 kelvins higher than what it would be if the planet had no atmosphere at all.

TEST YOUR UNDERSTANDING OF SECTION 42.2 A rotating diatomic molecule emits a photon when it makes a transition from level (n, l) to level $(n - 1, l - 1)$. If the value of l increases but n is unchanged, does the wavelength of the emitted photon (i) increase, (ii) decrease, or (iii) remain unchanged?

ANSWER

(ii) Figure 42.5 shows that the difference in energy between adjacent rotational levels increases with increasing l. Hence, as l increases, the energy E of the emitted photon increases and the wavelength $\lambda = hc/E$ decreases.

42.3 STRUCTURE OF SOLIDS

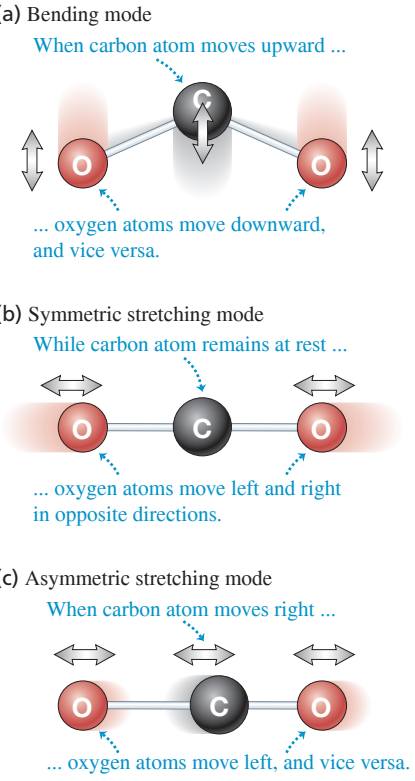
The term *condensed matter* includes both solids and liquids. In both states, the interactions between atoms or molecules are strong enough to give the material a definite volume that changes relatively little with applied stress. In condensed matter, adjacent atoms attract one another until their outer electron charge clouds begin to overlap significantly. Thus the distances between adjacent atoms in condensed matter are about the same as the diameters of the atoms themselves, typically 0.1 to 0.5 nm. Also, when we speak of the distances between atoms, we mean the center-to-center (nucleus-to-nucleus) distances.

Ordinarily, we think of a liquid as a material that can flow and of a solid as a material with a definite shape. However, if you heat a horizontal glass rod in the flame of a burner, you'll find that the rod begins to sag (flow) more and more easily as its temperature rises. Glass has no definite transition from solid to liquid, and no definite melting point. On this basis, we can consider glass at room temperature as being an extremely viscous liquid. Tar and butter show similar behavior.

What is the microscopic difference between materials like glass or butter and solids like ice or copper, which do have definite melting points? Ice and copper are examples of *crystalline solids* in which the atoms have *long-range order,* a recurring pattern of atomic positions that extends over many atoms. This pattern is called the *crystal structure*. In contrast, glass at room temperature is an example of an *amorphous* solid, one that has no long-range order but only *short-range order* (correlations between neighboring atoms or molecules). Liquids also have only short-range order. The boundaries between crystalline solid, amorphous solid, and liquid may be sometimes blurred. Some solids, crystalline when perfect, can form with so many imperfections in their structure that they have almost no long-range order. (Yet another kind of order is found in a *liquid crystal,* which is made up of rod-shaped or disc-shaped molecules of an organic compound. The positions of the molecules in the liquid are not fixed, but there is *orientational* order; the axes of the molecules tend to align with each other. This ordering can extend over a distance of many molecules.)

Nearly everything we know about crystal structure was learned from diffraction experiments with x rays, electrons, or neutrons. A typical distance between atoms is of the order of 0.1 nm. You can show that 12.4 keV x rays, 150 eV electrons, and 0.0818 eV neutrons all have wavelengths $\lambda = 0.1$ nm.

Figure **42.10** The carbon dioxide molecule can vibrate in three different modes. For clarity, the atoms are not shown to scale: The separation between atoms is actually comparable to their diameters.



BIO APPLICATION Using Crystals to Determine Protein Structure Protein molecules can form crystals, such as these crystals of insulin (a protein composed of 51 amino acids). All of the molecules within a single crystal of a protein have the same orientation; how the crystal diffracts x rays or neutrons depends on the shape and size of the molecules. By analyzing these diffraction patterns, scientists have deduced the molecular structures of more than 100,000 types of proteins.

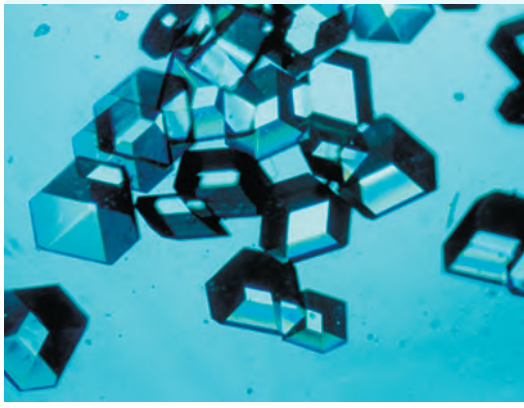
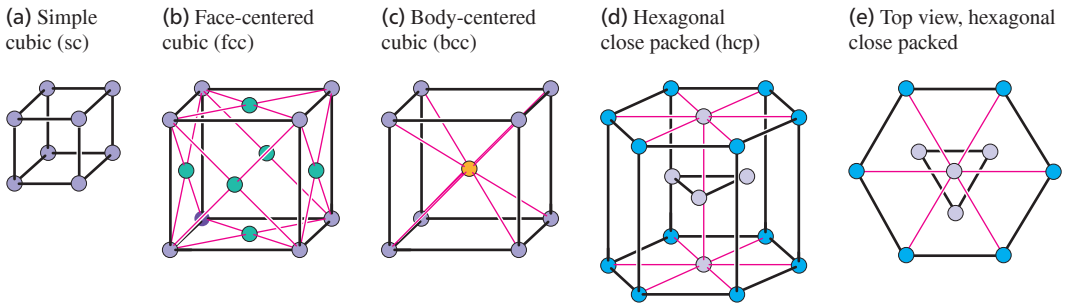


Figure 42.11 Portions of some common types of crystal lattices.

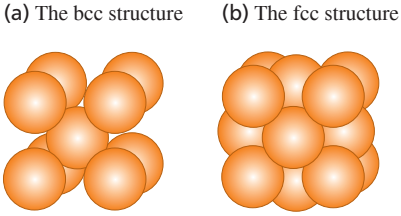


Crystal Lattices and Structures

An essential part of understanding crystals is the idea of a *crystal lattice,* which is a repeating pattern of mathematical points that extends throughout space. There are 14 general types of such patterns; **Fig. 42.11** shows small portions of some common examples. The *simple cubic lattice* (sc) has a lattice point at each corner of a cubic array (Fig. 42.11a). The *face-centered cubic lattice* (fcc) is like the simple cubic but with an additional lattice point at the center of each cube face (Fig. 42.11b). The *body-centered cubic lattice* (bcc) is like the simple cubic but with an additional point at the center of each cube (Fig. 42.11c). The *hexagonal close-packed lattice* (hcp) has layers of lattice points in hexagonal patterns, each hexagon made up of six equilateral triangles (Figs. 42.11d and 42.11e).

CAUTION **A perfect crystal lattice is infinitely large** Figure 42.11 shows just enough lattice points so that you can easily visualize the pattern; the lattice, a mathematical abstraction, extends throughout space. Thus the lattice points shown repeat endlessly in all directions.

Figure 42.12 (a) The bcc *structure* is composed of a bcc *lattice* with a basis of one atom for each lattice point. (b) The fcc *structure* is composed of an fcc *lattice* with a basis of one atom for each lattice point. These structures repeat precisely to make up perfect crystals.



In a crystal structure, a single atom or a group of atoms is associated with each lattice point. The group may contain the same or different kinds of atoms. This atom or group of atoms is called a *basis.* Thus a complete description of a crystal structure includes both the lattice and the basis. We initially consider *perfect crystals,* or *ideal single crystals,* in which the crystal structure extends uninterrupted throughout space.

The bcc and fcc structures are two common simple crystal structures. The alkali metals have a bcc structure—that is, a bcc lattice with a basis of one atom at each lattice point. Each atom in a bcc structure has eight nearest neighbors (**Fig. 42.12a**). The elements Al, Ca, Cu, Ag, and Au have an fcc structure—that is, an fcc lattice with a basis of one atom at each lattice point. Each atom in an fcc structure has 12 nearest neighbors (Fig. 42.12b).

Figure 42.13 Representation of part of the sodium chloride crystal structure. The distances between ions are exaggerated.

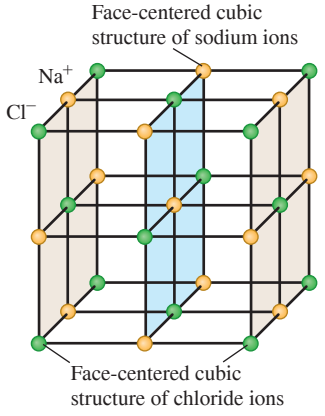
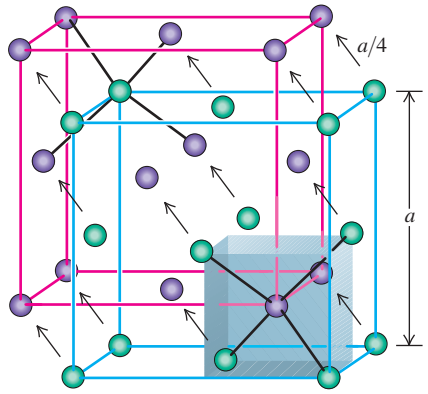


Figure 42.13 shows a representation of the structure of sodium chloride (NaCl, ordinary salt). It may look like a simple cubic structure, but it isn't. The sodium and chloride ions each form an fcc structure, so we can think roughly of the sodium chloride structure as being composed of two interpenetrating fcc structures. More correctly, the sodium chloride crystal structure of Fig. 42.13 has an fcc lattice with one chloride ion at each lattice point and one sodium ion half a cube length above it. That is, its basis consists of one chloride and one sodium ion.

Another example is the *diamond structure;* it's called that because it is the crystal structure of carbon in the diamond form. It's also the crystal structure of silicon, germanium, and gray tin. The diamond lattice is fcc; the basis consists of one atom at each lattice point and a second *identical* atom displaced a quarter of a cube length in each of the three cube-edge directions. **Figure 42.14** will help you visualize this. The shaded volume in Fig. 42.14 shows the bottom right front eighth of the basic cube; the four atoms at alternate corners of this cube are at the corners of a regular tetrahedron, and there is an additional atom at the center. Thus each atom in the diamond structure is at the center of a regular tetrahedron with four nearest-neighbor atoms at the corners.

In the diamond structure, both the purple and green spheres in Fig. 42.14 represent *identical* atoms—for example, both carbon or both silicon. In the cubic zinc sulfide structure, the purple spheres represent one type of atom and the green spheres represent a *different* type. For example, in zinc sulfide (ZnS) each zinc atom (purple in Fig. 42.14) is at the center of a regular tetrahedron with four sulfur atoms (green in Fig. 42.14) at its corners, and vice versa. Gallium arsenide (GaAs) and similar compounds have this same structure.

Figure **42.14** The diamond structure, shown as two interpenetrating face-centered cubic structures with distances between atoms exaggerated. Relative to the corresponding green atom, each purple atom is shifted up, back, and to the left by a distance $a/4$.



Bonding in Solids

The forces that are responsible for the regular arrangement of atoms in a crystal are the same as those involved in molecular bonds, plus one additional type discussed below. Not surprisingly, *ionic* and *covalent* molecular bonds are found in ionic and covalent crystals, respectively. The most familiar *ionic crystals* are the alkali halides, such as ordinary salt (NaCl). The positive sodium ions and the negative chloride ions occupy adjacent positions in the crystal (see Fig. 42.13). The attractive forces are the familiar Coulomb's-law forces between charged particles. These forces have no preferred direction, and the arrangement in which the material crystallizes is partly determined by the relative sizes of the two ions. Such a structure is *stable* in the sense that it has lower total energy than the separated ions (see the following example). The negative potential energies of pairs of opposite charges are greater in absolute value than the positive energies of pairs of like charges because the pairs of unlike charges are closer together, on average.

EXAMPLE 42.4 Potential energy of an ionic crystal

Imagine a one-dimensional ionic crystal consisting of a very large number of alternating positive and negative ions with charges e and $-e$, with equal spacing a along a line. Show that the total interaction potential energy is negative, which means that such a "crystal" is stable.

IDENTIFY and SET UP We treat each ion as a point charge and use our results from Section 23.1 for the electric potential energy of a collection of point charges. Equations (23.10) and (23.11) tell us to consider the electric potential energy U of each pair of charges. The total potential energy of the system is the sum of the values of U for every possible pair; we take the number of pairs to be infinite.

EXECUTE Let's pick an ion somewhere in the middle of the line and add the potential energies of its interactions with all the ions to one side of it. From Eq. (23.11), that sum is

$$\begin{aligned}\Sigma U &= -\frac{e^2}{4\pi\epsilon_0}\frac{1}{a} + \frac{e^2}{4\pi\epsilon_0}\frac{1}{2a} - \frac{e^2}{4\pi\epsilon_0}\frac{1}{3a} + \cdots \\ &= -\frac{e^2}{4\pi\epsilon_0 a}\left(1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \cdots\right)\end{aligned}$$

You may notice that the series in parentheses resembles the Taylor series for the function $\ln(1 + x)$:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

When $x = 1$ this becomes the series in parentheses above, so

$$\Sigma U = -\frac{e^2}{4\pi\epsilon_0 a}\ln 2$$

This is certainly a negative quantity. The atoms on the other side of the ion we're considering make an equal contribution to the potential energy. And if we include the potential energies of all pairs of atoms, the sum is certainly negative.

EVALUATE We conclude that this one-dimensional ionic "crystal" is stable: It has lower energy than the zero electric potential energy that is obtained when all the ions are infinitely far apart from each other.

KEYCONCEPT An ionic crystal is stable because the net electric potential energy of its ions is less than if the crystal were taken apart and the ions were moved far from each other.

Types of Crystals

Carbon, silicon, germanium, and tin in the diamond structure are simple examples of *covalent crystals*. These elements are in Group IV of the periodic table, meaning that each atom has four electrons in its outermost shell. Each atom forms a covalent bond with each of four adjacent atoms at the corners of a tetrahedron (Fig. 42.14). These bonds are strongly directional because of the asymmetric electron distributions dictated by the exclusion principle (see Fig. 42.3), and the result is the tetrahedral diamond structure.

A third crystal type, less directly related to the chemical bond than are ionic or covalent crystals, is the **metallic crystal.** In this structure, one or more of the outermost

Figure 42.15 In a metallic solid, one or more electrons are detached from each atom and are free to wander around the crystal, forming an "electron gas." The wave functions for these electrons extend over many atoms. The positive ions vibrate around fixed locations in the crystal.

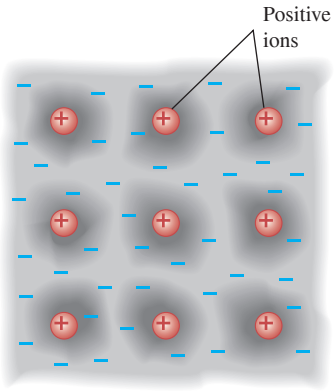
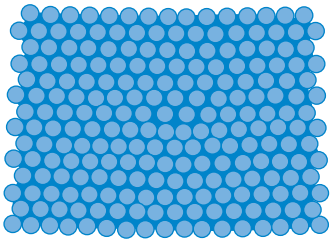


Figure 42.16 An edge dislocation in two dimensions. In three dimensions an edge dislocation would look like an extra plane of atoms slipped partway into the crystal.



You can see the irregularity most easily by viewing the figure from various directions at a grazing angle with the page.

Figure 42.17 The concept of energy bands was first developed by the Swiss-American physicist Felix Bloch (1905–1983) in his doctoral thesis. Our modern understanding of electrical conductivity stems from that landmark work. Bloch's work in nuclear physics brought him (along with Edward Purcell) the 1952 Nobel Prize in physics.



electrons in each atom become detached from the parent atom (leaving a positive ion). The detached electrons are free to move through the crystal and are not localized near the ion from which they originated. So we can picture a metallic crystal as an array of positive ions immersed in a sea of freed electrons whose attraction for the positive ions holds the crystal together (**Fig. 42.15**).

This sea of electrons, which gives metals their high electrical and thermal conductivities, has many of the properties of a gas. Indeed, we speak of the *electron-gas model* of metallic solids. The simplest version of this model is the *free-electron model,* which ignores interactions with the ions completely (except at the surface). We'll return to this model in Section 42.5.

In a metallic crystal the freed electrons are shared among *many* atoms. This gives a bonding that is neither localized nor strongly directional. The crystal structure is determined primarily by considerations of *close packing*—that is, the maximum number of atoms that can fit into a given volume. The two most common metallic crystal lattices are the face-centered cubic and hexagonal close-packed (see Figs. 42.11b, 42.11d, and 42.11e). In structures composed of these lattices with a basis of one atom, each atom has 12 nearest neighbors.

Hydrogen bonds and van der Waals forces also play a role in the structure of some solids. In polyethylene and similar polymers, covalent bonding of atoms forms long-chain molecules, and hydrogen bonding forms cross-links between adjacent chains. In solid water, both hydrogen bonds and van der Waals forces are significant in determining the crystal structures of ice.

Our discussion has centered on perfect crystals. Real crystals show a variety of departures from this idealized structure. Materials are often *polycrystalline,* composed of many small single crystals bonded together at *grain boundaries.* There may be *point defects* within a crystal: *Interstitial* atoms may occur in places where they do not belong, and there may be *vacancies,* positions that should be occupied by an atom but are not. A point defect of interest in semiconductors, which we'll discuss in Section 42.6, is the *substitutional impurity,* a foreign atom replacing a regular atom (for example, arsenic in a silicon crystal).

There are several basic types of extended defects called *dislocations.* One type is the *edge dislocation,* shown schematically in **Fig. 42.16**, in which one plane of atoms slips relative to another. The mechanical properties of metallic crystals are influenced strongly by the presence of dislocations. The ductility and malleability of some metals depend on the presence of dislocations that can move through the crystal during plastic deformations. The biggest extended defect of all, present in *all* real crystals, is the surface of the material with its dangling bonds and abrupt change in potential energy.

TEST YOUR UNDERSTANDING OF SECTION 42.3 If a is the distance in an NaCl crystal from an Na^+ ion to one of its nearest-neighbor Cl^- ions, what is the distance from an Na^+ ion to one of its *next-to-nearest*-neighbor Cl^- ions? (i) $a\sqrt{2}$; (ii) $a\sqrt{3}$; (iii) $2a$; (iv) none of these.

ANSWER (ii) In Fig. 42.13 let a be the distance between adjacent Na^+ and Cl^- ions. This figure shows that the Cl^- ion that is the next nearest neighbor to an Na^+ ion is on the opposite corner of a cube of side a. The distance between these two ions is $\sqrt{a^2 + a^2 + a^2} = \sqrt{3a^2} = a\sqrt{3}$.

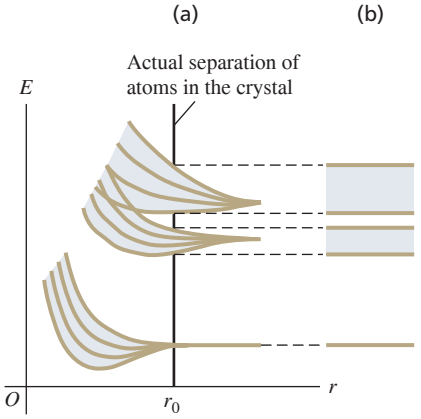
42.4 ENERGY BANDS

The **energy-band** concept, introduced in 1928 (**Fig. 42.17**), is a great help in understanding several properties of solids. To introduce the idea, suppose we have a large number N of identical atoms, far enough apart that their interactions are negligible. Every atom has the same energy-level diagram. We can draw an energy-level diagram for the *entire system.* It looks just like the diagram for a single atom, but the exclusion principle, applied to the entire system, permits each state to be occupied by N electrons (one per atom) instead of just one.

Now we begin to push the atoms uniformly closer together. Because of the electrical interactions and the exclusion principle, the wave functions begin to distort, especially

those of the outer, or *valence,* electrons. The corresponding energies also shift, some upward and some downward, by varying amounts, as the valence electron wave functions become less localized and extend over more and more atoms. (The inner electrons in an atom are affected much less by nearby atoms than are the valence electrons, and their energy levels remain relatively sharp.) Thus the valence states that formerly gave the *system* a state with a sharp energy level that could accommodate N electrons now give a *band* containing N closely spaced levels (**Fig. 42.18**). Ordinarily, N is somewhere near the order of Avogadro's number (10^{24}), so we can accurately treat the levels as forming a *continuous* distribution of energies within a band. Between adjacent energy bands are gaps where there are *no* allowed energy levels.

Figure **42.18** Origin of energy bands in a solid. (a) As the distance r between atoms decreases, the energy levels spread into bands. The vertical line at r_0 shows the actual atomic spacing in the crystal. (b) Symbolic representation of energy bands.



Insulators, Semiconductors, and Conductors

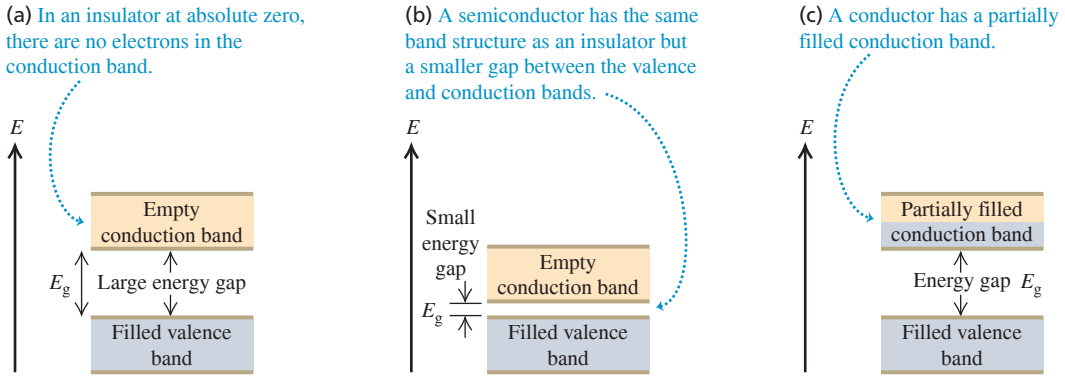
The nature of the energy bands determines whether the material is an electrical insulator, a semiconductor, or a conductor. In particular, what matters are the extent to which the states in each band are occupied and the spacing, called the *band gap* or *energy gap,* between adjacent bands.

In an *insulator* at absolute zero temperature, the highest band that is completely filled, called the **valence band,** is also the highest band that has *any* electrons in it. The next higher band, called the **conduction band,** is completely empty; there are no electrons in its states (**Fig. 42.19a**). Imagine what happens if an electric field is applied to a material of this kind. To move in response to the field, an electron would have to go into a different quantum state with a slightly different energy. It can't do that, however, because all the neighboring states are already occupied. The only way such an electron can move is to jump across the energy gap into the conduction band, where there are plenty of nearby unoccupied states. At any temperature above absolute zero there is some probability this jump can happen, because an electron can gain energy from thermal motion. In an insulator, however, the energy gap between the valence and conduction bands can be 5 eV or more, and that much thermal energy is not ordinarily available. Hence little or no current flows in response to an applied electric field, and the electrical conductivity (Section 25.2) is low. The thermal conductivity (Section 17.6), which also depends on mobile electrons, is likewise low.

We saw in Section 24.4 that an insulator becomes a conductor if it is subjected to a large enough electric field; this is called *dielectric breakdown*. If the electric field is of the order of 10^{10} V/m, there is a potential difference of a few volts over a distance comparable to atomic sizes. In this case the field can do enough work on a valence electron to boost it across the energy gap and into the conduction band. (In practice dielectric breakdown occurs for fields much less than 10^{10} V/m, because imperfections in the structure of an insulator provide some more accessible energy states *within* the energy gap.)

As in an insulator, a *semiconductor* at absolute zero has an empty conduction band above the full valence band. The difference is that in a semiconductor the energy gap

Figure **42.19** Three types of energy-band structure.



between these bands is relatively small and electrons can more readily jump into the conduction band (Fig. 42.19b). As the temperature of a semiconductor increases, the population in the conduction band increases very rapidly, as does the electrical conductivity. For example, in a semiconductor near room temperature with an energy gap of 1 eV, the number of conduction electrons doubles when the temperature rises by just 10°C. We'll use the concept of energy bands to explore semiconductors in more depth in Section 42.6.

In a *conductor* such as a metal, there are electrons in the conduction band even at absolute zero (Fig. 42.19c). The metal sodium is an example. An analysis of the atomic energy-level diagram for sodium (see Fig. 39.19a) shows that for an isolated sodium atom, the six lowest excited states (all 3*p* states) are about 2.1 eV above the two 3*s* ground states. In solid sodium, however, the atoms are so close together that the 3*s* and 3*p* *bands* spread out and overlap into a single band. Each sodium atom contributes one electron to the band, leaving an Na^+ ion behind. Each atom also contributes eight *states* to that band (two 3*s*, six 3*p*), so the band is only one-eighth occupied. We call this structure a *conduction* band because it is only partially occupied. Electrons near the top of the filled portion of the band have many adjacent unoccupied states available, and they can easily gain or lose small amounts of energy in response to an applied electric field. Therefore these electrons are mobile, giving solid sodium its high electrical and thermal conductivity. A similar description applies to other conducting materials.

EXAMPLE 42.5 Photoconductivity in germanium

At room temperature, pure germanium has an almost completely filled valence band separated by a 0.67 eV gap from an almost completely empty conduction band. It is a poor electrical conductor, but its conductivity increases greatly when it is irradiated with electromagnetic waves of a certain maximum wavelength. What is that wavelength?

IDENTIFY and SET UP The conductivity of a semiconductor increases greatly when electrons are excited from the valence band into the conduction band. In germanium, the excitation occurs when an electron absorbs a photon with an energy of at least $E_{\text{min}} = 0.67\text{ eV}$. From $E = hc/\lambda$, the *maximum* wavelength λ_{max} (our target variable) corresponds to this *minimum* photon energy.

EXECUTE The wavelength of a photon with energy $E_{\text{min}} = 0.67\text{ eV}$ is

$$\lambda_{\text{max}} = \frac{hc}{E_{\text{min}}} = \frac{(4.136 \times 10^{-15}\text{ eV}\cdot\text{s})(3.00 \times 10^{8}\text{ m/s})}{0.67\text{ eV}}$$
$$= 1.9 \times 10^{-6}\text{ m} = 1.9\ \mu\text{m} = 1900\text{ nm}$$

EVALUATE This wavelength is in the infrared part of the spectrum, so visible-light photons (which have shorter wavelength and higher energy) will also induce conductivity in germanium. As we'll see in Section 42.7, semiconductor crystals are widely used as photocells and for many other applications.

KEYCONCEPT The electron energy levels in a semiconductor form a valence band that is essentially filled, plus a conduction band that is empty and separated from the valence band by a small energy gap. The conductivity of a semiconductor can be increased by using photons to excite electrons from the valence band into the conduction band. To do this, the photon energy must be at least as great as the band gap; lower-energy photons will not be absorbed.

TEST YOUR UNDERSTANDING OF SECTION 42.4 One type of thermometer works by measuring the temperature-dependent electrical resistivity of a sample. Which of the following types of material displays the greatest change in resistivity for a given temperature change? (i) Insulator; (ii) semiconductor; (iii) conductor.

ANSWER (ii) A small temperature change causes a substantial increase in the population of electrons in a semiconductor's conduction band and a comparably substantial increase in conductivity. The conductivity of conductors and insulators varies more gradually with temperature.

42.5 FREE-ELECTRON MODEL OF METALS

Studying the energy states of electrons in metals can give us a lot of insight into their electrical and magnetic properties, the electron contributions to heat capacities, and other behavior. As we discussed in Section 42.3, one of the distinguishing features of a metal is that one or more valence electrons are detached from their home atom and can move freely within the metal, with wave functions that extend over many atoms.

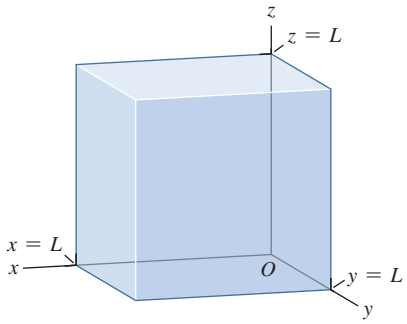
The **free-electron model** assumes that these electrons don't interact at all with the ions or with each other, but that there are infinite potential-energy barriers at the surfaces. The idea is that a typical electron moves so rapidly within the metal that it "sees" the effect of the ions and other electrons as a uniform potential-energy function, whose value we can choose to be zero.

We can represent the surfaces of the metal by the same cubical box that we analyzed in Section 41.2 (the three-dimensional version of the particle in a box studied in Section 40.2). If the box has sides of length L (**Fig. 42.20**), the energies of the stationary states (quantum states of definite energy) are

$$E_{n_X,n_Y,n_Z} = \frac{(n_X^2 + n_Y^2 + n_Z^2)\pi^2\hbar^2}{2mL^2} \qquad \begin{matrix} (n_X = 1, 2, 3, \ldots; \\ n_Y = 1, 2, 3, \ldots; \\ n_Z = 1, 2, 3, \ldots) \end{matrix} \qquad (42.10)$$

Each state is labeled by the three positive-integer quantum numbers (n_X, n_Y, n_Z).

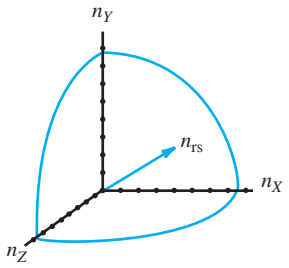
Figure 42.20 A cubical box with side length L. We studied this three-dimensional version of the infinite square well in Section 41.2. The energy levels for a particle in this box are given by Eq. (42.10).



Density of States

Later we'll need to know the *number* dn of quantum states that have energies in a given range dE. The number of states per unit energy range dn/dE is called the **density of states,** denoted by $g(E)$. We'll begin by working out an expression for $g(E)$. Think of a three-dimensional space with coordinates (n_X, n_Y, n_Z) (**Fig. 42.21**). The radius n_{rs} of a sphere centered at the origin in that space is $n_{rs} = (n_X^2 + n_Y^2 + n_Z^2)^{1/2}$. Each point with integer coordinates in that space represents one spatial quantum state. Thus each point corresponds to one unit of volume in the space, and the total number of points with integer coordinates inside a sphere equals the volume of the sphere, $\frac{4}{3}\pi n_{rs}^3$. Because all our n's are positive, we must take only one *octant* of the sphere, with $\frac{1}{8}$ the total volume, or $(\frac{1}{8})(\frac{4}{3}\pi n_{rs}^3) = \frac{1}{6}\pi n_{rs}^3$. The particles are electrons, so each point corresponds to *two* states with opposite spin components $(m_s = \pm\frac{1}{2})$, and the total number n of electron states corresponding to points inside the octant is twice $\frac{1}{6}\pi n_{rs}^3$, or

Figure 42.21 The allowed values of n_X, n_Y, and n_Z are positive integers for the electron states in the free-electron gas model. Including spin, there are two states for each unit volume in n space.



$$n = \frac{\pi n_{rs}^3}{3} \qquad (42.11)$$

The energy E of states at the surface of the sphere can be expressed in terms of n_{rs}. Equation (42.10) becomes

$$E = \frac{n_{rs}^2\pi^2\hbar^2}{2mL^2} \qquad (42.12)$$

We can combine Eqs. (42.11) and (42.12) to get a relationship between E and n that doesn't contain n_{rs}. We'll leave the details for you to work out; the total number of states with energies of E or less is

$$n = \frac{(2m)^{3/2}VE^{3/2}}{3\pi^2\hbar^3} \qquad (42.13)$$

where $V = L^3$ is the volume of the box.

To get the number of states dn in an energy interval dE, we treat n and E as continuous variables and take differentials of both sides of Eq. (42.13):

$$dn = \frac{(2m)^{3/2}VE^{1/2}}{2\pi^2\hbar^3}\,dE \qquad (42.14)$$

The density of states $g(E)$ is equal to dn/dE, so from Eq. (42.14) we get

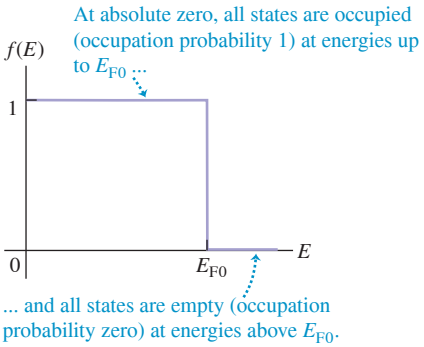
Density of states, free-electron model:

$$g(E) = \frac{(2m)^{3/2}V}{2\pi^2\hbar^3}E^{1/2} \tag{42.15}$$

($g(E)$: Number of states per unit energy range near E; m: Electron mass; V: Volume; E: Electron energy; $\hbar$: Planck's constant divided by 2π)

Fermi–Dirac Distribution

Figure 42.22 The probability distribution for occupation of free-electron energy states at absolute zero.



Let's now see how the electrons are distributed among the various quantum states at any given temperature. The Maxwell–Boltzmann distribution states that the average number of particles in a state of energy E is proportional to $e^{-E/kT}$ (see Sections 18.5 and 39.4). However, there are two reasons why it wouldn't be right to use the Maxwell–Boltzmann distribution. The first reason is the exclusion principle. At absolute zero the Maxwell–Boltzmann function predicts that *all* the electrons would go into the two ground states of the system, with $n_X = n_Y = n_Z = 1$ and $m_s = \pm\frac{1}{2}$. But the exclusion principle allows only one electron in each state. At absolute zero the electrons can fill up the lowest *available* states, but they cannot *all* go into the lowest states. Thus at absolute zero the distribution function is as shown in **Fig. 42.22**. All states with energies E less than some value E_{F0} are occupied, so the occupation probability $f(E) = 1$; all states with energies greater than this value are unoccupied, so $f(E) = 0$.

The second reason we can't use the Maxwell–Boltzmann distribution is more subtle. That distribution assumes that the particles are *distinguishable*. But, as we discussed in Section 41.8, electrons are *indistinguishable;* it's impossible to "tag" electrons to know which is which. If one electron is in state A and the other is in state B, there's no way to tell whether electron 1 is in state A and electron 2 is in state B, or electron 1 is in state B and electron 2 is in state A.

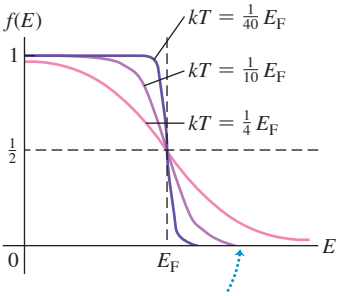
The statistical distribution function that emerges from the exclusion principle and the indistinguishability requirement is called (after its inventors) the **Fermi–Dirac distribution.** This distribution gives the probability $f(E)$ that at temperature T, a particular state of energy E is occupied by an electron:

Fermi–Dirac distribution:

$$f(E) = \frac{1}{e^{(E - E_F)/kT} + 1} \tag{42.16}$$

($f(E)$: Probability that a given state is occupied by an electron; T: Absolute temperature; E: Energy of state; E_F: Fermi energy; k: Boltzmann constant)

Figure 42.23 Graphs of the Fermi–Dirac distribution function for various values of kT, assuming that the Fermi energy E_F is independent of the temperature T.



The energy E_F is called the **Fermi energy** or the *Fermi level;* we'll discuss its significance below. We use E_{F0} for its value at absolute zero $(T = 0)$ and E_F for other temperatures. We can accurately let $E_F = E_{F0}$ for metals because the Fermi energy does not change much with temperature for solid conductors. However, it is not safe to assume that $E_F = E_{F0}$ for semiconductors, in which the Fermi energy usually does change with temperature.

Figure 42.23 shows graphs of Eq. (42.16) for three temperatures. The $kT = \frac{1}{40}E_F$ curve shows that for $kT \ll E_F$, $f(E)$ approaches the shape of the absolute-zero distribution function shown in Fig. 42.22.

Note that when $E = E_F$, the exponent $(E - E_F)/kT$ in Eq. (42.16) is zero, so $f(E) = f(E_F) = \frac{1}{2}$ for any temperature T. That is, the probability is $\frac{1}{2}$ that a state at the Fermi energy contains an electron. This means that at $E = E_F$, half the states are filled (and half are empty). For states with $E < E_F$, the exponent is negative and the occupation probability $f(E)$ is greater than $\frac{1}{2}$; for states with $E > E_F$, $f(E)$ is less than $\frac{1}{2}$ and approaches zero for E much larger than kT.

EXAMPLE 42.6 Probabilities in the free-electron model

WITH √ARIATION PROBLEMS

For free electrons in a solid at temperature T, at what energy is the probability that a particular state is occupied equal to (a) 0.01 and (b) 0.99?

IDENTIFY and SET UP This problem asks us to explore the Fermi–Dirac distribution. Equation (42.16) gives the occupation probability $f(E)$ for a given energy E. If we solve this equation for E, we get an expression for the energy that corresponds to a given occupation probability—which is just what we need.

EXECUTE Using Eq. (42.16), you can show that

$$E = E_F + kT \ln\left(\frac{1}{f(E)} - 1\right)$$

(a) When $f(E) = 0.01$,

$$E = E_F + kT \ln\left(\frac{1}{0.01} - 1\right) = E_F + 4.6kT$$

The probability that a state $4.6kT$ above the Fermi level is occupied is only 0.01, or 1%.

(b) When $f(E) = 0.99$,

$$E = E_F + kT \ln\left(\frac{1}{0.99} - 1\right) = E_F - 4.6kT$$

The probability that a state $4.6kT$ below the Fermi level is occupied is 0.99, or 99%.

EVALUATE At very low temperatures, $4.6kT$ is much less than E_F. Then the occupation probability of levels even slightly below E_F is nearly 1 (100%), and that for levels even slightly above E_F is nearly zero (see Fig. 42.23). In general, if the probability is P that a state with an energy ΔE *above* E_F is occupied, then the probability is $1 - P$ that a state ΔE *below* E_F is occupied. We leave the proof to you.

KEYCONCEPT In the free-electron model of a metal, the Fermi–Dirac distribution gives the probability that a state of a given energy E is occupied at a temperature T. The probability is greater than $\frac{1}{2}$ for energies less than the Fermi energy E_F and less than $\frac{1}{2}$ for energies greater than E_F.

Electron Concentration and Fermi Energy

Equation (42.16) gives the probability that any specific state with energy E is occupied at a temperature T. To get the actual number of electrons in any energy range dE, we have to multiply this probability by the number dn of states in that range $g(E)\,dE$. Thus the number dN of electrons with energies in the range dE is

$$dN = g(E)f(E)\,dE = \frac{(2m)^{3/2}VE^{1/2}}{2\pi^2\hbar^3}\frac{1}{e^{(E-E_F)/kT} + 1}\,dE \qquad (42.17)$$

The Fermi energy E_F is determined by the total number N of electrons; at any temperature the electron states are filled up to a point at which all electrons are accommodated. At absolute zero there is a simple relationship between E_{F0} and N. *All* states below E_{F0} are filled; in Eq. (42.13) we set n equal to the total number of electrons N and E to the Fermi energy at absolute zero E_{F0}:

$$N = \frac{(2m)^{3/2}V{E_{F0}}^{3/2}}{3\pi^2\hbar^3} \qquad (42.18)$$

Solving Eq. (42.18) for E_{F0}, we get

$$E_{F0} = \frac{3^{2/3}\pi^{4/3}\hbar^2}{2m}\left(\frac{N}{V}\right)^{2/3} \qquad (42.19)$$

The quantity N/V is the number of free electrons per unit volume. It is called the *electron concentration* and is usually denoted by n.

If we replace N/V with n, Eq. (42.19) becomes

$$E_{F0} = \frac{3^{2/3}\pi^{4/3}\hbar^2 n^{2/3}}{2m} \qquad (42.20)$$

CAUTION **Electron concentration and number of electrons** Don't confuse the electron concentration n with any quantum number n. Furthermore, the number of states is *not* in general the same as the total number of electrons N.

EXAMPLE 42.7 The Fermi energy in copper

WITH √ARIATION PROBLEMS

At low temperatures, copper has a free-electron concentration $n = 8.45 \times 10^{28}\ \text{m}^{-3}$. Using the free-electron model, find the Fermi energy for solid copper, and find the speed of an electron with a kinetic energy equal to the Fermi energy.

IDENTIFY and SET UP This problem uses the relationship between Fermi energy and free-electron concentration. Because copper is a solid conductor, its Fermi energy changes very little with temperature and we can use the expression for the Fermi energy at absolute zero, Eq. (42.20). We'll use the nonrelativistic formula $E_F = \frac{1}{2}mv_F^2$ to find the *Fermi speed* v_F that corresponds to kinetic energy E_F.

EXECUTE Using the given value of n, we solve for E_F and v_F:

$$E_F = \frac{3^{2/3}\pi^{4/3}(1.055 \times 10^{-34}\ \text{J}\cdot\text{s})^2(8.45 \times 10^{28}\ \text{m}^{-3})^{2/3}}{2(9.11 \times 10^{-31}\ \text{kg})}$$

$$= 1.126 \times 10^{-18}\ \text{J} = 7.03\ \text{eV}$$

$$v_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2(1.126 \times 10^{-18}\ \text{J})}{9.11 \times 10^{-31}\ \text{kg}}} = 1.57 \times 10^6\ \text{m/s}$$

EVALUATE Our values of E_F and v_F are within the ranges of typical values for metals, 1.6–14 eV and 0.8–2.2 $\times 10^6$ m/s, respectively. Note that the calculated Fermi speed is far less than the speed of light $c = 3.00 \times 10^8$ m/s, which justifies our use of the nonrelativistic formula $\frac{1}{2}mv_F^2 = E_F$.

Our calculated Fermi energy is much larger than kT at ordinary temperatures. (At room temperature $T = 20°\text{C} = 293$ K, the quantity kT equals $(1.381 \times 10^{-23}\ \text{J/K})(293\ \text{K}) = 4.04 \times 10^{-21}\ \text{J} = 0.0254$ eV.) So it is a good approximation to take almost all the states below E_F as completely full and almost all those above E_F as completely empty (see Fig. 42.22).

We can also use Eq. (42.15) to find $g(E)$ if E and V are known. You can show that if $E = 7.03$ eV and $V = 1\ \text{cm}^3$, $g(E)$ is about 2×10^{22} states/eV. This huge number shows why we were justified in treating n and E as continuous variables in our density-of-states derivation.

KEYCONCEPT The Fermi energy in the free-electron model of a metal depends on the electron concentration (number of electrons per unit volume). In a typical metal at room temperature T, the Fermi energy is much greater than kT.

Average Free-Electron Energy

We can calculate the *average* free-electron energy in a metal at absolute zero by using the same ideas that we used to find E_{F0}. From Eq. (42.17) the number dN of electrons with energies in the range dE is $g(E)f(E)\,dE$. The energy of these electrons is $E\,dN = Eg(E)f(E)\,dE$. At absolute zero we substitute $f(E) = 1$ from $E = 0$ to $E = E_{F0}$ and $f(E) = 0$ for all other energies. Therefore the total energy E_{tot} of all the N electrons is

$$E_{tot} = \int_0^{E_{F0}} Eg(E)(1)\,dE + \int_{E_{F0}}^{\infty} Eg(E)(0)\,dE = \int_0^{E_{F0}} Eg(E)\,dE$$

The simplest way to evaluate this expression is to compare Eqs. (42.15) and (42.19). You'll see that

$$g(E) = \frac{3NE^{1/2}}{2E_{F0}^{3/2}}$$

Substituting this expression into the integral and using $E_{av} = E_{tot}/N$, we get

$$E_{av} = \frac{3}{2E_{F0}^{3/2}} \int_0^{E_{F0}} E^{3/2}\,dE = \tfrac{3}{5}E_{F0} \qquad (42.21)$$

At absolute zero the average free-electron energy equals $\frac{3}{5}$ of the Fermi energy.

EXAMPLE 42.8 Free-electron gas versus ideal gas

(a) Find the average energy of the free electrons in copper at absolute zero (see Example 42.7). (b) What would be the average kinetic energy of electrons if they behaved like an ideal gas at room temperature, 20°C (see Section 18.3)? What would be the speed of an electron with this kinetic energy? Compare these ideal-gas values with the (correct) free-electron values.

IDENTIFY and SET UP Free electrons in a metal behave like a kind of gas. In part (a) we use Eq. (42.21) to determine the average kinetic energy of free electrons in terms of the Fermi energy at absolute zero, which we know for copper from Example 42.7. In part (b) we treat electrons as an ideal gas at room temperature: Eq. (18.16) then gives the average kinetic energy per electron as $E_{av} = \frac{3}{2}kT$, and $E_{av} = \frac{1}{2}mv^2$ gives the corresponding electron speed v.

EXECUTE (a) From Example 42.7, the Fermi energy in copper at absolute zero is 1.126×10^{-18} J = 7.03 eV. According to Eq. (42.21), the average energy is $\frac{3}{5}$ of this, or 6.76×10^{-19} J = 4.22 eV.

(b) In Example 42.7 we found that $kT = 4.04 \times 10^{-21}$ J = 0.0254 eV at room temperature $T = 20°C = 293$ K. If electrons behaved like an ideal gas at this temperature, the average kinetic energy per electron would be $\frac{3}{2}$ of this, or 6.07×10^{-21} J = 0.0379 eV. The speed of an electron with this kinetic energy is

$$v = \sqrt{\frac{2E_{av}}{m}} = \sqrt{\frac{2(6.07 \times 10^{-21}\,\text{J})}{9.11 \times 10^{-31}\,\text{kg}}} = 1.15 \times 10^5\ \text{m/s}$$

EVALUATE The ideal-gas model predicts an average energy that is about 1% of the value given by the free-electron model, and a speed that is about 7% of the free-electron Fermi speed $v_F = 1.57 \times 10^6$ m/s that we found in Example 42.7. Thus temperature plays a *very* small role in determining the properties of electrons in metals; their average energies are determined almost entirely by the exclusion principle.

A similar analysis allows us to determine the contributions of electrons to the heat capacities of a solid metal. If there is one conduction electron per atom, the principle of equipartition of energy (see Section 18.4) would predict that the kinetic energies of these electrons contribute $3R/2$ to the molar heat capacity at constant volume C_V. But when kT is much smaller than E_F, which is usually the situation in metals, only those few electrons near the Fermi level can find empty states and change energy appreciably when the temperature changes. The number of such electrons is proportional to kT/E_F, so we expect that the electron molar heat capacity at constant volume is proportional to $(kT/E_F)(3R/2) = (3kT/2E_F)R$. A more detailed analysis shows that the actual electron contribution to C_V for a solid metal is $(\pi^2 kT/2E_F)R$, not far from our prediction. You can verify that if $T = 293$ K and $E_F = 7.03$ eV, the electron contribution to C_V is $0.018R$, which is only 1.2% of the (incorrect) $3R/2$ prediction of the equipartition principle. Because the electron contribution is so small, the overall heat capacity of most solid metals is due primarily to vibration of the atoms in the crystal structure (see Fig. 18.18 in Section 18.4).

KEYCONCEPT The exclusion principle, not the effects of temperature, is the dominant reason for the distribution of electron energies in a metal at room temperature.

TEST YOUR UNDERSTANDING OF SECTION 42.5 An ideal gas obeys the relationship $pV = nRT$ (see Section 18.1): For a given volume V and a number of moles n, as the temperature T decreases, the pressure p decreases proportionately and tends to zero as T approaches absolute zero. Is this also true of the free-electron gas in a solid metal?

ANSWER no The kinetic-molecular model of an ideal gas (Section 18.3) shows that the gas pressure is proportional to the average translational kinetic energy E_{av} of the particles that make up the gas. In a classical ideal gas, E_{av} is directly proportional to the average temperature T, so the pressure decreases as T decreases. In a free-electron gas, the average kinetic energy per electron is *not* related simply to T; as Example 42.8 shows, for the free-electron gas in a metal, E_{av} is almost completely a consequence of the exclusion principle at room temperature and colder. Hence the pressure of a free-electron gas in a solid metal does *not* change appreciably between room temperature and absolute zero.

42.6 SEMICONDUCTORS

A **semiconductor** has an electrical resistivity that is intermediate between those of good conductors and those of good insulators. The tremendous importance of semiconductors in present-day electronics stems in part from the fact that their electrical properties are very sensitive to very small concentrations of impurities. We'll discuss the basic concepts, using the semiconductor elements silicon (Si) and germanium (Ge) as examples.

Silicon and germanium are in Group IV of the periodic table. Both have four electrons in the outermost atomic subshells ($3s^2 3p^2$ for silicon, $4s^2 4p^2$ for germanium), and both crystallize in the covalently bonded diamond structure discussed in Section 42.3 (see Fig. 42.14). Because all four of the outer electrons are involved in the bonding, at absolute zero the band structure (see Section 42.4) has a completely empty conduction band (see Fig. 42.19b). As we discussed in Section 42.4, at very low temperatures electrons cannot jump from the filled valence band into the conduction band. This property makes these materials insulators at very low temperatures; their electrons have no nearby states available into which they can move in response to an applied electric field.

However, in semiconductors the energy gap E_g between the valence and conduction bands is small in comparison to the gap of 5 eV or more for many insulators; room-temperature values are 1.12 eV for silicon and only 0.67 eV for germanium. Thus even at room temperature a substantial number of electrons can gain enough energy to jump the gap to the conduction band, where they are dissociated from their parent atoms and are free to move about the crystal. The number of these electrons increases rapidly with temperature.

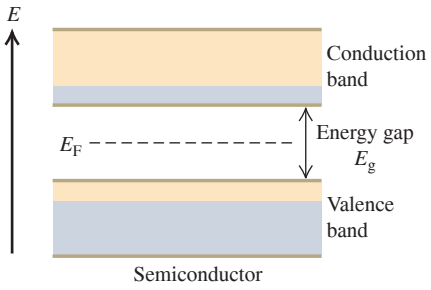
EXAMPLE 42.9 Jumping a band gap

WITH √ARIATION PROBLEMS

Consider a material with the band structure described above, with its Fermi energy in the middle of the gap (**Fig. 42.24**). Find the probability that a state at the bottom of the conduction band is occupied at $T = 300$ K, and compare that with the probability at $T = 310$ K, for band gaps of (a) 0.200 eV; (b) 1.00 eV; (c) 5.00 eV.

IDENTIFY and SET UP The Fermi–Dirac distribution function gives the probability that a state of energy E is occupied at temperature T. Figure 42.24 shows that the state of interest at the bottom of the conduction band has an energy $E = E_F + E_g/2$ that is greater than the Fermi energy E_F, with $E - E_F = E_g/2$. Figure 42.23 shows that the higher the temperature, the larger the fraction of electrons with energies greater than the Fermi energy.

Figure **42.24** Band structure of a semiconductor. At absolute zero a completely filled valence band is separated by a narrow energy gap E_g of 1 eV or so from a completely empty conduction band. At ordinary temperatures, a number of electrons are excited to the conduction band.



EXECUTE (a) When $E_g = 0.200$ eV,

$$\frac{E - E_F}{kT} = \frac{E_g}{2kT} = \frac{0.100\ \text{eV}}{(8.617 \times 10^{-5}\ \text{eV/K})(300\ \text{K})} = 3.87$$

$$f(E) = \frac{1}{e^{3.87} + 1} = 0.0205$$

For $T = 310$ K, the exponent is 3.74 and $f(E) = 0.0231$, a 13% increase in probability for a temperature rise of 10 K.

(b) For $E_g = 1.00$ eV, both exponents are five times as large as in part (a), namely 19.3 and 18.7; the values of $f(E)$ are 4.0×10^{-9} and 7.4×10^{-9}. In this case the (low) probability nearly doubles with a temperature rise of 10 K.

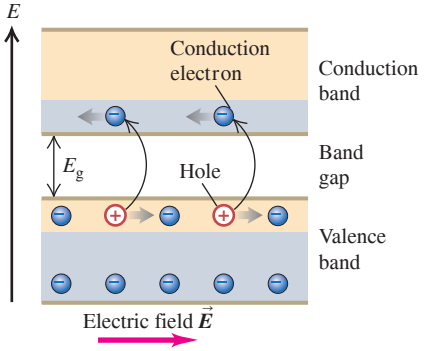
(c) For $E_g = 5.0$ eV, the exponents are 96.7 and 93.6; the values of $f(E)$ are 1.0×10^{-42} and 2.3×10^{-41}. The (extremely low) probability increases by a factor of 23 for a 10 K temperature rise.

EVALUATE This example illustrates two important points. First, the probability of finding an electron in a state at the bottom of the conduction band is extremely sensitive to the width of the band gap. At room temperature, the probability is about 2% for a 0.200 eV gap, a few in a thousand million for a 1.00 eV gap, and essentially zero for a 5.00 eV gap. (Pure diamond, with a 5.47 eV band gap, has essentially no electrons in the conduction band and is an excellent insulator.) Second, for any given band gap the probability depends strongly on temperature, and even more strongly for large gaps than for small ones.

KEYCONCEPT The properties of the Fermi–Dirac distribution explain why the number of electrons in the conduction band of a semiconductor is sensitive to both the temperature and the size of the band gap.

In principle, we could continue the calculation in Example 42.9 to find the actual density $n = N/V$ of electrons in the conduction band at any temperature. To do this, we would have to evaluate the integral $\int g(E)f(E)\,dE$ from the bottom of the conduction band to its top. [To do this we would need to know the density of states function $g(E)$. This function for a semiconductor is more complicated than the free-electron model expression given by Eq. (42.15).] Once we know n, we can *begin* to determine the resistivity of the material (and its temperature dependence) by using the analysis of Section 25.2, which you may want to review. But next we'll see that the electrons in the conduction band don't tell the whole story about conduction in semiconductors.

Figure **42.25** Motion of electrons in the conduction band and of holes in the valence band of a semiconductor under the action of an applied electric field $\vec{E}$.



Holes

When an electron is removed from a covalent bond, it leaves a vacancy behind. An electron from a neighboring atom can move into this vacancy, leaving the neighbor with the vacancy. In this way the vacancy, called a **hole,** can travel through the material and serve as an additional current carrier. It's like describing the motion of a bubble in a liquid. In a pure, or *intrinsic,* semiconductor, valence-band holes and conduction-band electrons are always present in equal numbers. When an electric field is applied, they move in opposite directions (**Fig. 42.25**). Thus a hole in the valence band behaves like a positively charged particle, even though the moving charges in that band are electrons. The conductivity that we just described for a pure semiconductor is called *intrinsic conductivity.* Another kind of conductivity, to be discussed in the next subsection, is due to impurities.

An analogy helps to picture conduction in an intrinsic semiconductor. The valence band at absolute zero is like a floor of a parking garage that's filled bumper to bumper with cars (which represent electrons). No cars can move because there is nowhere for them

to go. But if one car is moved to the vacant floor above, it can move freely, just as electrons can move freely in the conduction band. Also, the empty space that it leaves permits cars to move on the nearly filled floor, thereby moving the empty space just as holes move in the normally filled valence band.

Impurities

Suppose we mix into melted germanium (Ge, $Z = 32$) a small amount of arsenic (As, $Z = 33$), the next element after germanium in the periodic table. We then allow the mixture to cool and crystallize. This deliberate addition of impurity elements is called *doping*. Arsenic is in Group V; it has *five* valence electrons compared to the four valence electrons of germanium. When one of these five valence electrons is removed from an arsenic atom, the remaining electron structure is essentially identical to that of germanium. The only difference is that it is smaller; the arsenic nucleus has a charge of $+33e$ rather than $+32e$, and it pulls the electrons in a little more. An arsenic atom can comfortably take the place of a germanium atom as a substitutional impurity. Four of its five valence electrons form the necessary nearest-neighbor covalent bonds.

The fifth valence electron in arsenic is very loosely bound (**Fig. 42.26a**); it doesn't participate in the covalent bonds, and it is screened from the nuclear charge of $+33e$ by the 32 electrons, leaving a net effective charge of about $+e$. We might guess that the binding energy would be of the same order of magnitude as the energy of the $n = 4$ level in hydrogen—that is, $(\frac{1}{4})^2(13.6\text{ eV}) = 0.85\text{ eV}$. In fact, it is much smaller than this, only about 0.01 eV, because the electron probability distribution actually extends over many atomic diameters and the polarization of intervening atoms provides additional screening.

The energy level of this fifth electron corresponds in the band picture to an isolated energy level lying in the gap, about 0.01 eV below the bottom of the conduction band (Fig. 42.26b). This level is called a *donor level,* and the impurity atom that is responsible for it is simply called a *donor.* All Group V elements, including N, P, As, Sb, and Bi, can serve as donors. At room temperature, kT is about 0.025 eV. This is substantially greater than 0.01 eV, so at ordinary temperatures, most electrons can gain enough energy to jump from donor levels into the conduction band, where they are free to wander through the material. The remaining ionized donor stays at its site in the structure and does not participate in conduction.

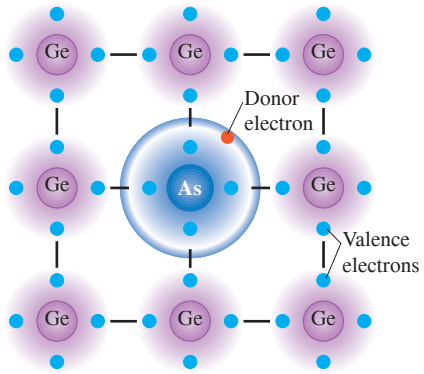
Example 42.9 shows that at ordinary temperatures and with a band gap of 1.0 eV, only a very small fraction (of the order of 10^{-9}) of the states at the bottom of the conduction band in a pure semiconductor contain electrons to participate in intrinsic conductivity. Thus we expect the conductivity of such a semiconductor to be about 10^{-9} as great as that of good metallic conductors, and measurements bear out this prediction. However, a concentration of donors as small as one part in 10^8 can increase the conductivity so drastically that conduction due to impurities becomes by far the dominant mechanism. In this case the conductivity is due almost entirely to *negative* charge (electron) motion. We call the material an ***n*-type semiconductor,** with *n*-type impurities.

Adding atoms of an element in Group III (B, Al, Ga, In, Tl), with only *three* valence electrons, has an analogous effect. An example is gallium (Ga, $Z = 31$); as a substitutional impurity in germanium, the gallium atom would like to form four covalent bonds, but it has only three outer electrons. It can, however, steal an electron from a neighboring germanium atom to complete the required four covalent bonds (**Fig. 42.27a**, next page). The resulting atom has the same electron configuration as germanium but is somewhat larger because gallium's nuclear charge is smaller, $+31e$ instead of $+32e$.

This theft leaves the neighboring atom with a *hole,* or missing electron. The hole acts as a positive charge that can move through the crystal just as with intrinsic conductivity. The stolen electron is bound to the gallium atom in a level called an *acceptor level* about 0.01 eV above the top of the valence band (Fig. 42.27b). The gallium atom, called an *acceptor,* thus accepts an electron to complete its desire for four covalent bonds. This extra electron gives the previously neutral gallium atom a net charge of $-e$. The resulting gallium ion is

Figure **42.26** An *n*-type semiconductor: germanium (Ge) with an arsenic (As) impurity.

(a) A donor (*n*-type) impurity atom has a fifth valence electron that does not participate in the covalent bonding and is very loosely bound.



(b) Energy-band diagram for an *n*-type semiconductor at a low temperature. One donor electron has been excited from the donor levels into the conduction band.

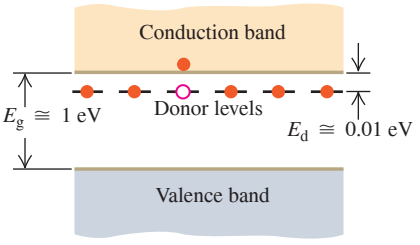
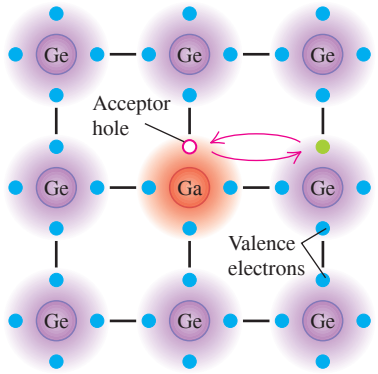


Figure **42.27** A *p*-type semiconductor: germanium (Ge) with a gallium (Ga) impurity.

(a) An acceptor (*p*-type) impurity atom has only three valence electrons, so it can borrow an electron from a neighboring atom. The resulting hole is free to move about the crystal.



(b) Energy-band diagram for a *p*-type semiconductor at a low temperature. One acceptor level has accepted an electron from the valence band, leaving a hole behind.

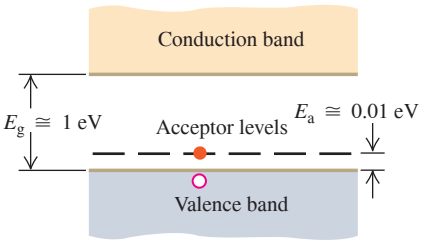
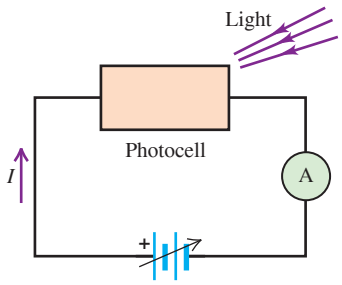


Figure **42.28** A semiconductor photocell in a circuit. The more intense the light falling on the photocell, the greater the conductivity of the photocell and the greater the current measured by the ammeter (A).



not free to move. In a semiconductor that is doped with acceptors, we consider the conductivity to be almost entirely due to *positive* charge (hole) motion. We call the material a ***p*-type semiconductor,** with *p*-type impurities. Some semiconductors are doped with *both* *n*- and *p*-type impurities. Such materials are called *compensated* semiconductors.

CAUTION **The meaning of "*p*-type" and "*n*-type"** It's a common misconception that a *p*-type semiconductor has a net positive charge and an *n*-type semiconductor has a net negative charge. In fact, both types of semiconductors have a net *zero* charge because they are made up of neutral atoms. Instead, "*p*-type" means that the majority of the mobile carriers of charge within the semiconductor are positive (holes), and "*n*-type" means that the majority of mobile carriers are negative (electrons).

We can verify the assertion that the current in *n*- and *p*-type semiconductors really *is* carried by electrons and holes, respectively, by using the Hall effect (see Section 27.9). The sign of the Hall emf is opposite in the two cases. Hall-effect devices constructed from semiconductor materials are used in probes to measure magnetic fields and the currents that cause those fields.

TEST YOUR UNDERSTANDING OF SECTION 42.6 Would there be any advantage to adding *n*-type or *p*-type impurities to copper?

ANSWER no Pure copper is already an excellent conductor since it has a partially filled conduction band (Fig. 42.19c). Furthermore, copper forms a metallic crystal (Fig. 42.15) as opposed to the covalent crystals of silicon or germanium, so the scheme of using an impurity to donate or accept an electron does not work for copper. In fact, adding impurities to copper *decreases* the conductivity because an impurity tends to scatter electrons, impeding the flow of current.

42.7 SEMICONDUCTOR DEVICES

Semiconductor devices play an indispensable role in contemporary electronics. In the early days of radio and television, transmitting and receiving equipment relied on vacuum tubes, but these have been replaced by solid-state devices, including transistors, diodes, integrated circuits, and other semiconductor devices. All modern consumer electronic devices use semiconductor devices of various kinds.

One simple semiconductor device is the *photocell* (**Fig. 42.28**). When a thin slab of semiconductor is irradiated with an electromagnetic wave whose photons have at least as much energy as the band gap between the valence and conduction bands, an electron in the valence band can absorb a photon and jump to the conduction band, where it and the hole it left behind contribute to the conductivity (see Example 42.5 in Section 42.4). The conductivity therefore increases with wave intensity, thus increasing the current I in the photocell circuit of Fig. 42.28. Hence the ammeter reading indicates the intensity of the light.

Detectors for charged particles operate on the same principle. An external circuit applies a voltage across a semiconductor. An energetic charged particle passing through the semiconductor collides inelastically with valence electrons, exciting them from the valence to the conduction band and creating pairs of holes and conduction electrons. The conductivity increases momentarily, causing a pulse of current in the external circuit. Solid-state detectors are widely used in nuclear and high-energy physics research.

The *p*-*n* Junction

In many semiconductor devices the essential principle is the fact that the conductivity of the material is controlled by impurity concentrations, which can be varied within wide limits from one region of a device to another. An example is the ***p*-*n* junction** at the boundary between one region of a semiconductor with *p*-type impurities and another region containing *n*-type impurities. One way of fabricating a *p*-*n* junction is to deposit some *n*-type material on the *very* clean surface of some *p*-type material. (We can't just stick *p*- and *n*-type pieces together and expect the junction to work properly because of the impossibility of matching their surfaces at the atomic level.)

When a *p*-*n* junction is connected to an external circuit, as in **Fig. 42.29a**, and the potential difference $V_p - V_n = V$ across the junction is varied, the current I varies as shown in Fig. 42.29b. In striking contrast to the symmetric behavior of resistors that obey Ohm's law and give a straight line on an *I*–*V* graph, a *p*-*n* junction conducts much more readily in the direction from *p* to *n* than the reverse. Such a (mostly) one-way device is called a **diode rectifier.** Later we'll discuss a simple model of *p*-*n* junction behavior that predicts a current–voltage relationship in the form

Current through a *p*-*n* junction: $I = I_S(e^{eV/kT} - 1)$ (42.22)

Saturation current; Voltage; Absolute temperature; Boltzmann constant; $e = 2.71828\ldots$; Magnitude of electron charge

CAUTION **Two different uses of *e*** In $e^{eV/kT}$ the base of the exponent also uses the symbol e, standing for the base of the natural logarithms, 2.71828.... This e is quite different from $e = 1.602 \times 10^{-19}$ C in the exponent.

Equation (42.22) is valid for both positive and negative values of V; note that V and I always have the same sign. As V becomes very negative, I approaches the value $-I_S$. The magnitude I_S (always positive) is called the *saturation current.*

Currents Through a *p*-*n* Junction

We can understand the behavior of a *p*-*n* junction diode qualitatively on the basis of the mechanisms for conductivity in the two regions. Suppose, as in Fig. 42.29a, you connect the positive terminal of the battery to the *p* region and the negative terminal to the *n* region. Then the *p* region is at higher potential than the *n* region, corresponding to positive V in Eq. (42.22), and the resulting electric field is in the direction *p* to *n*. This is called the *forward* direction, and the positive potential difference is called *forward bias*. Holes, plentiful in the *p* region, flow easily across the junction into the *n* region, and free electrons, plentiful in the *n* region, easily flow into the *p* region; these movements of charge constitute a *forward* current. Connecting the battery with the opposite polarity gives *reverse bias,* and the field tends to push electrons from *p* to *n* and holes from *n* to *p*. But there are very few free electrons in the *p* region and very few holes in the *n* region. As a result, the current in the *reverse* direction is much smaller than that with the same potential difference in the forward direction.

Suppose you have a box with a barrier separating the left and right sides: You fill the left side with oxygen gas and the right side with nitrogen gas. What happens if the barrier leaks? Oxygen diffuses to the right, and nitrogen diffuses to the left. A similar diffusion occurs across a *p*-*n* junction. First consider the equilibrium situation with no applied voltage (**Fig. 42.30**). The many holes in the *p* region act like a hole gas that diffuses across the junction into the *n* region. Once there, the holes recombine with some of the many free electrons. Similarly, electrons diffuse from the *n* region to the *p* region and fall into some of the many holes there. The hole and electron diffusion currents lead to a

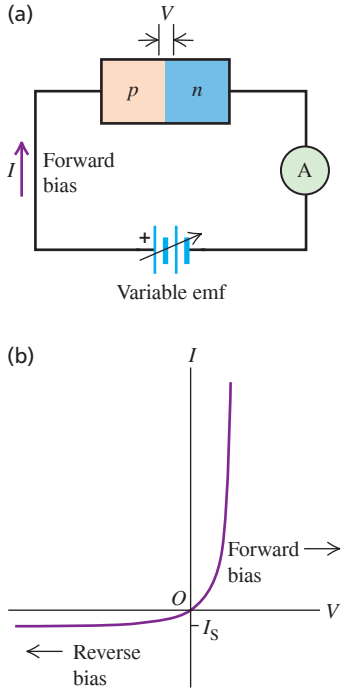


Figure **42.29** (a) A semiconductor *p*-*n* junction in a circuit. (b) Graph showing the asymmetric current–voltage relationship. The curve is described by Eq. (42.22).

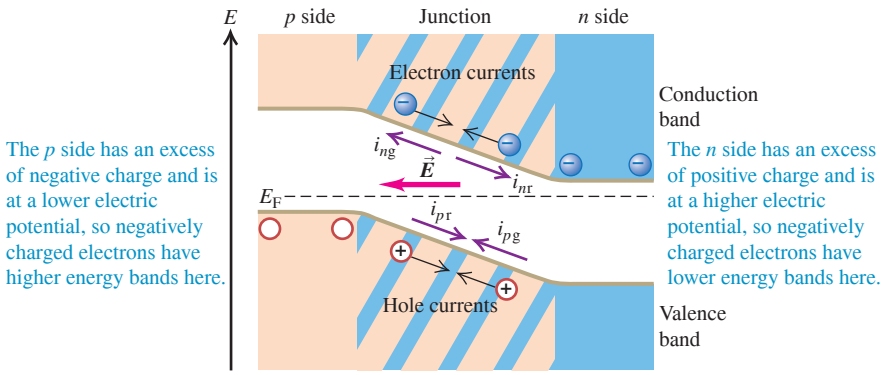


Figure **42.30** A *p*-*n* junction in equilibrium, with no externally applied field or potential difference. The generation currents (subscript g) and recombination currents (subscript r) exactly balance. The Fermi energy E_F is the same on both sides of the junction. The excess positive and negative charges on the *n* and *p* sides produce an electric field $\vec{E}$ in the direction shown.

net positive charge in the n region and a net negative charge in the p region, causing an electric field in the direction from n to p at the junction. The potential energy associated with this field raises the electron energy levels in the p region relative to the same levels in the n region.

There are four currents across the junction, as shown. The diffusion processes lead to *recombination currents* of holes and electrons, labeled i_{pr} and i_{nr} in Fig. 42.30. At the same time, electron–hole pairs are generated in the junction region by thermal excitation. The electric field described above sweeps these electrons and holes out of the junction; electrons are swept opposite the field to the n side, and holes are swept in the same direction as the field to the p side. The corresponding currents, called *generation currents,* are labeled i_{pg} and i_{ng}. At equilibrium the magnitudes of the generation and recombination currents are equal:

$$|i_{pg}| = |i_{pr}| \quad \text{and} \quad |i_{ng}| = |i_{nr}| \tag{42.23}$$

In thermal equilibrium the Fermi energy is the same at each point across the junction.

Now we apply a forward bias—that is, a positive potential difference V across the junction. A forward bias *decreases* the electric field in the junction region. It also decreases the difference between the energy levels on the p and n sides (**Fig. 42.31**) by an amount $\Delta E = -eV$. It becomes easier for the electrons in the n region to climb the potential-energy hill and diffuse into the p region and for the holes in the p region to diffuse into the n region. This effect increases both recombination currents by the Maxwell–Boltzmann factor $e^{-\Delta E/kT} = e^{eV/kT}$. (We don't have to use the Fermi–Dirac distribution because most of the available states for the diffusing electrons and holes are empty, so the exclusion principle has little effect.) The generation currents don't change appreciably, so the net hole current is

$$\begin{aligned} i_{p\text{tot}} &= i_{pr} - |i_{pg}| \\ &= |i_{pg}|e^{eV/kT} - |i_{pg}| \\ &= |i_{pg}|(e^{eV/kT} - 1) \end{aligned} \tag{42.24}$$

The net electron current $i_{n\text{tot}}$ is given by a similar expression, so the total current $I = i_{p\text{tot}} + i_{n\text{tot}}$ is

$$I = I_S(e^{eV/kT} - 1) \tag{42.25}$$

in agreement with Eq. (42.22). We can repeat this entire discussion for reverse bias (negative V and I) with the same result. Therefore Eq. (42.22) is valid for both positive and negative values.

Several effects make the behavior of practical p-n junction diodes more complex than this simple analysis predicts. One effect, *avalanche breakdown,* occurs under large reverse bias. The electric field in the junction is so great that the carriers can gain enough energy between collisions to create electron–hole pairs during inelastic collisions. The electrons and holes then gain energy and collide to form more pairs, and so on. (A similar effect occurs in dielectric breakdown in insulators, discussed in Section 42.4.)

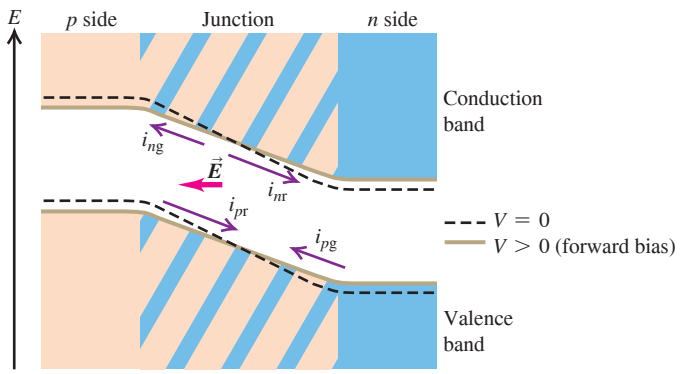
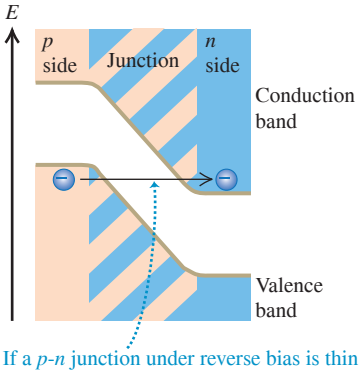


Figure **42.31** A p-n junction under forward-bias conditions. The potential difference between p and n regions is reduced, as is the electric field within the junction. The recombination currents increase but the generation currents are nearly constant, causing a net current from left to right. (Compare Fig. 42.30.)

A second type of breakdown begins when the reverse bias becomes large enough that the top of the valence band in the *p* region is just higher in energy than the bottom of the conduction band in the *n* region (**Fig. 42.32**). If the junction region is thin enough, the probability becomes large that electrons can *tunnel* from the valence band of the *p* region to the conduction band of the *n* region. This process is called *Zener breakdown*. It occurs in Zener diodes, which are used for voltage regulation and protection against voltage surges.

Figure **42.32** Under reverse-bias conditions the potential-energy difference between the *p* and *n* sides of a junction is greater than at equilibrium. If this difference is great enough, the bottom of the conduction band on the *n* side may actually be below the top of the valence band on the *p* side.

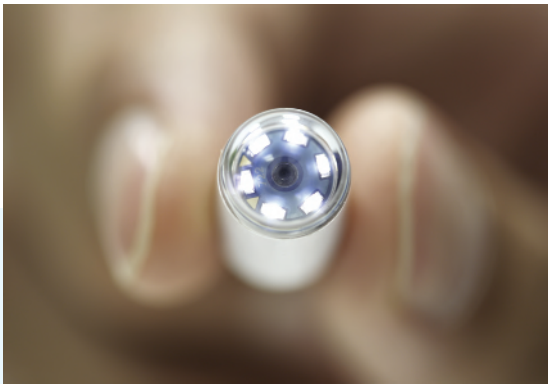


If a *p-n* junction under reverse bias is thin enough, electrons can tunnel from the valence band to the conduction band (a process called Zener breakdown).

Semiconductor Devices and Light

A *light-emitting diode (LED)* is a *p-n* junction diode that emits light. When the junction is forward biased, many holes are pushed from their *p* region to the junction region, and many electrons are pushed from their *n* region to the junction region. In the junction region the electrons fall into holes (recombine). In recombining, the electron can emit a photon with energy approximately equal to the band gap. This energy (and therefore the photon wavelength and the color of the light) can be varied by using materials with different band gaps. Light-emitting diodes are very energy-efficient light sources and have many applications, including automobile lamps, traffic signals, and flat-screen displays.

The reverse process is called the *photovoltaic effect*. Here the material absorbs photons, and electron–hole pairs are created. Pairs that are created in the *p-n* junction, or close enough to migrate to it without recombining, are separated by the electric field we described above that sweeps the electrons to the *n* side and the holes to the *p* side. We can connect this device to an external circuit, where it becomes a source of emf and power. Such a device is often called a *solar cell,* although sunlight isn't required. *Any* light with photon energies greater than the band gap will do. You might have a calculator powered by such cells. Production of low-cost photovoltaic cells for large-scale solar energy conversion is a very active field of research. The same basic physics is used in charge-coupled device (CCD) image detectors, digital cameras, and video cameras.



BIO APPLICATION Swallow This Semiconductor Device This tiny capsule—designed to be swallowed by a patient—contains a miniature camera with a CCD light detector, plus six LEDs to illuminate the subject. The capsule radios high-resolution images to an external recording unit as it passes painlessly through the patient's stomach and intestines. This technique makes it possible to examine the small intestine, which is not readily accessible with conventional endoscopy.

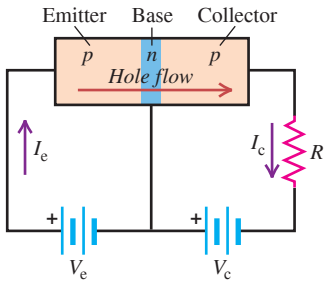
Transistors

A *bipolar junction transistor* includes two *p-n* junctions in a "sandwich" configuration, which may be either *p-n-p* or *n-p-n*. **Figure 42.33** shows such a *p-n-p* transistor. The three regions are called the emitter, base, and collector, as shown. When there is no current in the left loop of the circuit, there is only a very small current through the resistor *R* because the voltage across the base–collector junction is in the reverse direction. But when a forward bias is applied between emitter and base, as shown, most of the holes traveling from emitter to base travel *through* the base (which is typically both narrow and lightly doped) to the second junction, where they come under the influence of the collector-to-base potential difference and flow on through the collector to give an increased current to the resistor.

In this way the current in the collector circuit is *controlled* by the current in the emitter circuit. Furthermore, V_c may be considerably larger than V_e, so the *power* dissipated in *R* may be much larger than the power supplied to the emitter circuit by the battery V_e. Thus the device functions as a *power amplifier.* If the potential drop across *R* is greater than V_e, it may also be a voltage amplifier.

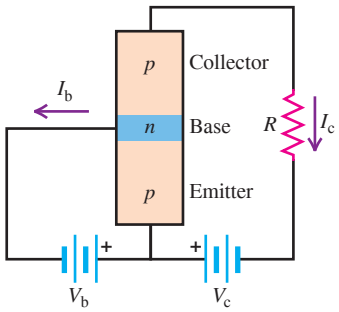
In this configuration the *base* is the common element between the "input" and "output" sides of the circuit. Another widely used arrangement is the *common-emitter* circuit, shown in **Fig. 42.34** (next page). In this circuit the current in the collector side of the circuit is much larger than that in the base side, and the result is current amplification.

Figure **42.33** Schematic diagram of a *p-n-p* transistor and circuit.



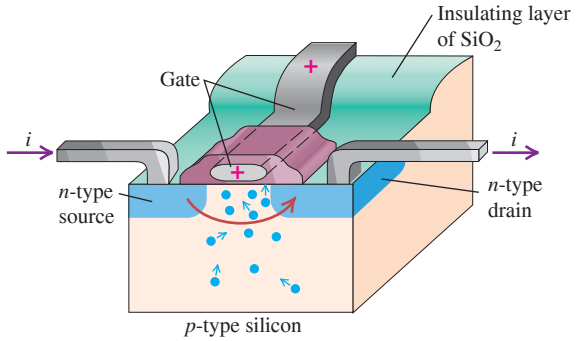
- When $V_e = 0$, the current is very small.
- When a potential V_e is applied between emitter and base, holes travel from the emitter to the base.
- When V_c is sufficiently large, most of the holes continue into the collector.

Figure **42.34** A common-emitter circuit.



- When $V_b = 0$, I_c is very small, and most of the voltage V_c appears across the base–collector junction.
- As V_b increases, the base–collector potential decreases, and more holes can diffuse into the collector; thus, I_c increases. Ordinarily, I_c is much larger than I_b.

Figure **42.35** A field-effect transistor. The current from source to drain is controlled by the potential difference between the source and the drain and by the charge on the gate; no current flows through the gate.

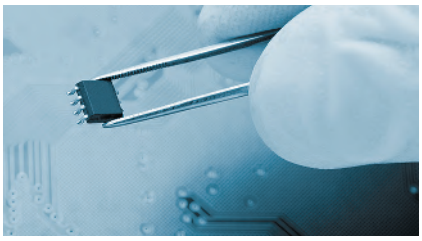


The *field-effect transistor* (**Fig. 42.35**) is an important type. In one variation a slab of *p*-type silicon is made with two *n*-type regions on the top, called the *source* and the *drain;* a metallic conductor is fastened to each. A third electrode called the *gate* is separated from the slab, source, and drain by an insulating layer of SiO_2. When there is no charge on the gate and a potential difference of either polarity is applied between the source and the drain, there is very little current because one of the *p*-*n* junctions is reverse biased.

Now we place a positive charge on the gate. With dimensions of the order of 10^{-6} m, it takes little charge to provide a substantial electric field. Thus there is very little current into or out of the gate. There aren't many free electrons in the *p*-type material, but there are some, and the effect of the field is to attract them toward the positive gate. The resulting greatly enhanced concentration of electrons near the gate (and between the two junctions) permits current to flow between the source and the drain. The current is very sensitive to the gate charge and potential, and the device functions as an amplifier. The device just described is called an *enhancement-type MOSFET* (metal-oxide-semiconductor field-effect transistor).

Integrated Circuits

Figure **42.36** An integrated circuit chip smaller than your fingertip can contain millions of transistors.



A further refinement in semiconductor technology is the *integrated circuit*. By successively depositing layers of material and etching patterns to define current paths, we can combine the functions of several MOSFETs, capacitors, and resistors on a single square of semiconductor material that may be only a few millimeters on a side. An elaboration of this idea leads to *large-scale integrated circuits*. The resulting integrated circuit chips are the heart of all pocket calculators, mobile devices, and present-day computers (**Fig. 42.36**).

The first semiconductor devices were invented in 1947. Since then, they have completely revolutionized the electronics industry through miniaturization, reliability, speed, energy usage, and cost. They have found applications in communications, computer systems, control systems, and many other areas. In transforming these areas, they have changed, and continue to change, human civilization itself.

TEST YOUR UNDERSTANDING OF SECTION 42.7 Suppose a negative charge is placed on the gate of the MOSFET shown in Fig. 42.35. Will a substantial current flow between the source and the drain?

ANSWER no A negative charge on the gate will repel, not attract, electrons in the *p*-type silicon. Hence the electron concentration in the region between the two *p*-*n* junctions will be made even smaller. With so few charge carriers present in this region, very little current will flow between the source and the drain.

42.8 SUPERCONDUCTIVITY

Superconductivity is the complete disappearance of all electrical resistance at low temperatures. We described this property at the end of Section 25.2 and the magnetic properties of type-I and type-II superconductors in Section 29.8. In this section we'll relate superconductivity to the structure and energy-band model of a solid.

Although superconductivity was discovered in 1911, it was not well understood on a theoretical basis until 1957. That year, the American physicists John Bardeen, Leon Cooper, and Robert Schrieffer published the theory of superconductivity, now called the BCS theory, that earned them the Nobel Prize in physics in 1972. (It was Bardeen's second; he shared his first for his work on the development of the transistor.) The key to the BCS theory is an interaction between *pairs* of conduction electrons, called *Cooper pairs,* caused by an interaction with the positive ions of the crystal. Here's a rough picture of what happens. A free electron exerts attractive forces on nearby positive ions, pulling them slightly closer together. The resulting slight concentration of positive charge then exerts an attractive force on another free electron with momentum opposite to the first. At ordinary temperatures this electron-pair interaction is very small in comparison to energies of thermal motion, but at very low temperatures it is significant.

Bound together this way, the pairs of electrons cannot *individually* gain or lose very small amounts of energy, as they would ordinarily be able to do in a partly filled conduction band. Their pairing gives an energy gap in the allowed electron quantum levels, and at low temperatures there is not enough collision energy to jump this gap. Therefore the electrons can move freely through the crystal without any energy exchange through collisions—that is, with zero resistance.

Since 1987 physicists have discovered a number of compounds that remain superconducting at temperatures above 77 K (the boiling point of liquid nitrogen). The original pairing mechanism of the BCS theory cannot explain the properties of these *high-temperature superconductors.* Instead, it appears that electrons in these materials form pairs due to magnetic interactions between their spins.

CHAPTER 42 SUMMARY

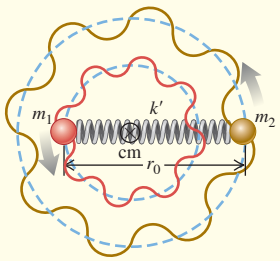
Molecular bonds and molecular spectra: The principal types of molecular bonds are ionic, covalent, van der Waals, and hydrogen bonds. In a diatomic molecule the rotational energy levels are given by Eq. (42.3), where I is the moment of inertia of the molecule, m_r is its reduced mass, and r_0 is the distance between the two atoms. The vibrational energy levels are given by Eq. (42.7), where k' is the effective force constant of the interatomic force. (See Examples 42.1–42.3.)

$$E_l = l(l+1)\frac{\hbar^2}{2I} \quad (l = 0, 1, 2, \ldots) \tag{42.3}$$

$$I = m_r r_0^2 \tag{42.6}$$

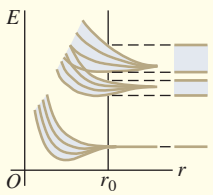
$$m_r = \frac{m_1 m_2}{m_1 + m_2} \tag{42.4}$$

$$E_n = (n + \tfrac{1}{2})\hbar\omega = (n + \tfrac{1}{2})\hbar\sqrt{\frac{k'}{m_r}} \quad (n = 0, 1, 2, \ldots) \tag{42.7}$$



Solids and energy bands: Interatomic bonds in solids are of the same types as in molecules plus one additional type, the metallic bond. Associating the basis with each lattice point gives the crystal structure. (See Example 42.4.)

When atoms are bound together in condensed matter, their outer energy levels spread out into bands. At absolute zero, insulators and conductors have a completely filled valence band separated by an energy gap from an empty conduction band. Conductors, including metals, have partially filled conduction bands. (See Example 42.5.)

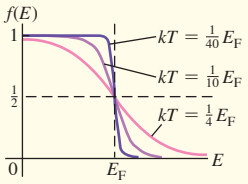


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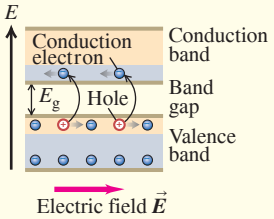
Free-electron model of metals: In the free-electron model of the behavior of conductors, the electrons are treated as completely free particles within the conductor. In this model the density of states is given by Eq. (42.15). The probability that an energy state of energy E is occupied is given by the Fermi–Dirac distribution, Eq. (42.16), which is a consequence of the exclusion principle. In Eq. (42.16), E_F is the Fermi energy. (See Examples 42.6–42.8.)

$$g(E) = \frac{(2m)^{3/2}V}{2\pi^2\hbar^3}E^{1/2} \qquad (42.15)$$

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \qquad (42.16)$$

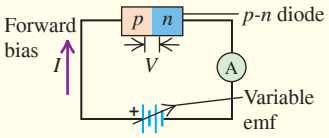


Semiconductors: A semiconductor has an energy gap of about 1 eV between its valence and conduction bands. Its electrical properties may be drastically changed by the addition of small concentrations of donor impurities, giving an n-type semiconductor, or acceptor impurities, giving a p-type semiconductor. (See Example 42.9.)



Semiconductor devices: Many semiconductor devices, including diodes, transistors, and integrated circuits, use one or more p-n junctions. The current–voltage relationship for an ideal p-n junction diode is given by Eq. (42.22).

$$I = I_S(e^{eV/kT} - 1) \qquad (42.22)$$



Chapter 42 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE √ARIATION PROBLEMS

Be sure to review EXAMPLES 42.2 and 42.3 (Section 42.2) before attempting these problems.

VP42.3.1 The two nuclei in the nitric oxide (NO) molecule are 0.1154 nm apart. The mass of the most common nitrogen atom is 2.326×10^{-26} kg, and the mass of the most common oxygen atom is 2.656×10^{-26} kg. Find (a) the reduced mass of the NO molecule, (b) the moment of inertia of the NO molecule, and (c) the energies, in meV, of the lowest three rotational energy levels of NO.

VP42.3.2 The $l = 2$ rotational level of the hydrogen chloride (HCl) molecule has energy 7.90 meV. The mass of the most common hydrogen atom is 1.674×10^{-27} kg, and the mass of the most common chlorine atom is 5.807×10^{-26} kg. Find (a) the moment of inertia of the HCl molecule, (b) the reduced mass of the HCl molecule, and (c) the distance between the H and Cl nuclei.

VP42.3.3 The mass of the most common silicon atom is 4.646×10^{-26} kg, and the mass of the most common oxygen atom is 2.656×10^{-26} kg. When a molecule of silicon monoxide (SiO) makes a transition between the $l = 1$ and $l = 0$ rotational levels, it emits a photon of wavelength 6.882 mm. Find (a) the moment of inertia of the SiO molecule, (b) the reduced mass of the SiO molecule, and (c) the distance between the Si and O nuclei.

VP42.3.4 A CO molecule is initially in the $n = 2$ vibrational level. If this molecule loses both vibrational and rotational energy and emits a photon, what are the photon wavelength and frequency if the initial angular momentum quantum number is (a) $l = 3$ and (b) $l = 2$?

Be sure to review EXAMPLES 42.6 and 42.7 (Section 42.5) before attempting these problems.

VP42.7.1 For free electrons in a solid with Fermi energy E_F and temperature T, find the energy for which the probability that a state at that energy is occupied is (a) 0.33 and (b) 0.90.

VP42.7.2 The Boltzmann constant is $k = 8.617 \times 10^{-5}$ eV/K. For a metallic solid at room temperature (293 K), what is the probability that an electron state is occupied if its energy is (a) 0.0250 eV below the Fermi level, (b) 0.0400 eV above the Fermi level, and (c) 0.100 eV above the Fermi level?

VP42.7.3 Silver contains 5.8×10^{28} free electrons per cubic meter. At absolute zero, what are (a) the Fermi energy (in J and eV) of silver, (b) the speed of an electron with this energy, and (c) the density of states (in states/J and states/eV) at the Fermi energy for a block of silver of volume 1.0 cm^3?

VP42.7.4 If kT is small compared to the Fermi energy at absolute zero, the Fermi energy at temperature T is essentially the same as at absolute zero. Use the results of the previous problem to find the electron energy, in eV, for which the probability that an electron state is occupied is (a) 0.92 and (b) 1.0×10^{-4} for silver at room temperature (293 K).

Be sure to review EXAMPLE 42.9 (Section 42.6) before attempting these problems.

VP42.9.1 For a certain semiconductor, the Fermi energy is in the middle of its band gap. If the temperature of the semiconductor is 285 K, find the probability that a state at the bottom of the conduction band is occupied if the band gap is (a) 0.500 eV and (b) 1.50 eV.

VP42.9.2 Find the ratio of the probability that a state at the bottom of the conduction band of a semiconductor is occupied at 315 K to the probability at 295 K if the band gap is (a) 0.400 eV and (b) 0.800 eV. Assume that the Fermi energy is in the middle of the band gap.

VP42.9.3 Consider a material whose Fermi energy is in the middle of its band gap. What band-gap width provides a probability of 0.00190 that a state at the bottom of the conduction band is occupied if the temperature of the material is (a) 305 K and (b) 325 K?

VP42.9.4 In a certain semiconductor, the Fermi energy lies above the top of the valence band by an amount equal to $\frac{3}{4}$ of the band gap. Find the probability that a state at the bottom of the conduction band is occupied if the band gap is 0.300 eV and the temperature is (a) 275 K and (b) 325 K.

BRIDGING PROBLEM Molecular Vibration and Semiconductor Band Gap

At 80 K, the band gap in the semiconductor indium antimonide (InSb) is 0.230 eV. A photon emitted by a hydrogen fluoride (HF) molecule undergoing a vibration-rotation transition from $(n = 1, l = 0)$ to $(n = 0, l = 1)$ is absorbed by an electron at the top of the valence band of InSb. (a) How far above the top of the band gap (in eV) is the final state of the electron? (b) What is the probability that the final state was already occupied? The vibration frequency for HF is 1.24×10^{14} Hz, the mass of a hydrogen atom is 1.67×10^{-27} kg, the mass of a fluorine atom is 3.15×10^{-26} kg, and the equilibrium distance between the two nuclei is 0.092 nm. Assume that the Fermi energy for InSb is in the middle of the gap.

SOLUTION GUIDE

IDENTIFY and SET UP

1. This problem involves what you learned about molecular transitions in Section 42.2, about the Fermi–Dirac distribution in Section 42.5, and about semiconductors in Section 42.6.
2. Equation (42.9) gives the combined vibrational-rotational energy in the initial and final molecular states. The difference between the initial and final molecular energies equals the energy E of the emitted photon, which is in turn equal to the energy gained by the InSb valence electron when it absorbs that photon. The probability that the final state is occupied is given by the Fermi–Dirac distribution, Eq. (42.16).

EXECUTE

3. Before you can use Eq. (42.9), you'll first need to use the data given to calculate the moment of inertia I and the quantity $\hbar\omega$ for the HF molecule. (*Hint:* Be careful not to confuse frequency f and angular frequency ω.)
4. Use your results from step 3 to calculate the initial and final energies of the HF molecule. (*Hint:* Does the vibrational energy increase or decrease? What about the rotational energy?)
5. Use your result from step 4 to find the energy imparted to the InSb electron. Determine the final energy of this electron relative to the bottom of the conduction band.
6. Use your result from step 5 to determine the probability that the InSb final state is already occupied.

EVALUATE

7. Is the molecular transition of the HF molecule allowed? Which is larger: the vibrational energy change or the rotational energy change?
8. Is it likely that the excited InSb electron will be blocked from entering a state in the conduction band?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q42.1 Van der Waals bonds occur in many molecules, but hydrogen bonds occur only with materials that contain hydrogen. Why is this type of bond unique to hydrogen?

Q42.2 The bonding of gallium arsenide (GaAs) is said to be 31% ionic and 69% covalent. Explain.

Q42.3 The H_2^+ molecule consists of two hydrogen nuclei and a single electron. What kind of molecular bond do you think holds this molecule together? Explain.

Q42.4 The moment of inertia for an axis through the center of mass of a diatomic molecule calculated from the wavelength emitted in an $l = 19 \rightarrow l = 18$ transition is different from the moment of inertia calculated from the wavelength of the photon emitted in an $l = 1 \rightarrow l = 0$ transition. Explain this difference. Which transition corresponds to the larger moment of inertia?

Q42.5 Analysis of the photon absorption spectrum of a diatomic molecule shows that the vibrational energy levels for small values of n are very nearly equally spaced but the levels for large n are not equally spaced. Discuss the reason for this observation. Do you expect the adjacent levels to move closer together or farther apart as n increases? Explain.

Q42.6 Discuss the differences between the rotational and vibrational energy levels of the deuterium ("heavy hydrogen") molecule D_2 and those of the ordinary hydrogen molecule H_2. A deuterium atom has twice the mass of an ordinary hydrogen atom.

Q42.7 Various organic molecules have been discovered in interstellar space. Why were these discoveries made with radio telescopes rather than optical telescopes?

Q42.8 The air you are breathing contains primarily nitrogen (N_2) and oxygen (O_2). Many of these molecules are in excited rotational energy levels $(l = 1, 2, 3, \ldots)$, but almost all of them are in the vibrational ground level $(n = 0)$. Explain this difference between the rotational and vibrational behaviors of the molecules.

Q42.9 In what ways do atoms in a diatomic molecule behave as though they were held together by a spring? In what ways is this a poor description of the interaction between the atoms?

Q42.10 Individual atoms have discrete energy levels, but certain solids (which are made up of only individual atoms) show energy bands and gaps. What causes the solids to behave so differently from the atoms of which they are composed?

Q42.11 What factors determine whether a material is a conductor of electricity or an insulator? Explain.

Q42.12 Ionic crystals are often transparent, whereas metallic crystals are always opaque. Why?

Q42.13 Speeds of molecules in a gas vary with temperature, whereas speeds of electrons in the conduction band of a metal are nearly independent of temperature. Why are these behaviors so different?

Q42.14 Use the band model to explain how it is possible for some materials to undergo a semiconductor-to-metal transition as the temperature or pressure varies.

Q42.15 An isolated zinc atom has a ground-state electron configuration of filled $1s$, $2s$, $2p$, $3s$, $3p$, and $4s$ subshells. How can zinc be a conductor if its valence subshell is full?

Q42.16 The assumptions of the *free-electron model* of metals may seem contrary to reason, since electrons exert powerful electric forces on each other. Give some reasons why these assumptions actually make physical sense.

Q42.17 Why are materials that are good thermal conductors also good electrical conductors? What kinds of problems does this pose for the design of appliances such as clothes irons and electric heaters? Are there materials that do not follow this general rule?

Q42.18 What is the essential characteristic for an element to serve as a donor impurity in a semiconductor such as Si or Ge? For it to serve as an acceptor impurity? Explain.

Q42.19 There are several methods for removing electrons from the surface of a semiconductor. Can holes be removed from the surface? Explain.

Q42.20 A student asserts that silicon and germanium become good insulators at very low temperatures and good conductors at very high temperatures. Do you agree? Explain your reasoning.

Q42.21 The electrical conductivities of most metals decrease gradually with increasing temperature, but the intrinsic conductivity of semiconductors always *increases* rapidly with increasing temperature. What causes the difference?

Q42.22 How could you make compensated silicon that has twice as many acceptors as donors?

Q42.23 The saturation current I_S for a p-n junction, Eq. (42.22), depends strongly on temperature. Explain why.

Q42.24 Why does tunneling limit the miniaturization of MOSFETs?

EXERCISES

Section 42.1 Types of Molecular Bonds

42.1 • **An Ionic Bond.** (a) Calculate the electric potential energy for a K^+ ion and a Br^- ion separated by a distance of 0.29 nm, the equilibrium separation in the KBr molecule. Treat the ions as point charges. (b) The ionization energy of the potassium atom is 4.3 eV. Atomic bromine has an electron affinity of 3.5 eV. Use these data and the results of part (a) to estimate the binding energy of the KBr molecule. Do you expect the actual binding energy to be higher or lower than your estimate? Explain your reasoning.

42.2 • During each of these processes, a photon of light is given up. In each process, what wavelength of light is given up, and in what part of the electromagnetic spectrum is that wavelength? (a) A molecule decreases its vibrational energy by 0.198 eV; (b) an atom decreases its energy by 7.80 eV; (c) a molecule decreases its rotational energy by 4.80×10^{-3} eV.

42.3 • For the H_2 molecule the equilibrium spacing of the two protons is 0.074 nm. The mass of a hydrogen atom is 1.67×10^{-27} kg. Calculate the wavelength of the photon emitted in the rotational transition $l = 3$ to $l = 1$.

Section 42.2 Molecular Spectra

42.4 •• The H_2 molecule has a moment of inertia of $4.6 \times 10^{-48}\ \text{kg} \cdot \text{m}^2$. What is the wavelength λ of the photon absorbed when H_2 makes a transition from the $l = 3$ to the $l = 4$ rotational level?

42.5 • A hypothetical NH molecule makes a rotational-level transition from $l = 3$ to $l = 1$ and gives off a photon of wavelength 1.710 nm in doing so. What is the separation between the two atoms in this molecule if we model them as point masses? The mass of hydrogen is 1.67×10^{-27} kg, and the mass of nitrogen is 2.33×10^{-26} kg.

42.6 • Two atoms of cesium (Cs) can form a Cs_2 molecule. The equilibrium distance between the nuclei in a Cs_2 molecule is 0.447 nm. Calculate the moment of inertia about an axis through the center of mass of the two nuclei and perpendicular to the line joining them. The mass of a cesium atom is 2.21×10^{-25} kg.

42.7 •• CP The rotational energy levels of CO are calculated in Example 42.2. If the energy of the rotating molecule is described by the classical expression $K = \frac{1}{2}I\omega^2$, for the $l = 1$ level what are (a) the angular speed of the rotating molecule; (b) the linear speed of each atom; (c) the rotational period (the time for one rotation)?

42.8 • The vibrational and rotational energies of the CO molecule are given by Eq. (42.9). Calculate the wavelength of the photon absorbed by CO in each of these vibration-rotation transitions: (a) $n = 0$, $l = 2 \rightarrow n = 1$, $l = 3$; (b) $n = 0$, $l = 3 \rightarrow n = 1$, $l = 2$; (c) $n = 0$, $l = 4 \rightarrow n = 1$, $l = 3$.

42.9 • A lithium atom has mass 1.17×10^{-26} kg, and a hydrogen atom has mass 1.67×10^{-27} kg. The equilibrium separation between the two nuclei in the LiH molecule is 0.159 nm. (a) What is the difference in energy between the $l = 3$ and $l = 4$ rotational levels? (b) What is the wavelength of the photon emitted in a transition from the $l = 4$ to the $l = 3$ level?

42.10 • If a sodium chloride (NaCl) molecule could undergo an $n \rightarrow n - 1$ vibrational transition with no change in rotational quantum number, a photon with wavelength 20.0 μm would be emitted. The mass of a sodium atom is 3.82×10^{-26} kg, and the mass of a chlorine atom is 5.81×10^{-26} kg. Calculate the force constant k' for the interatomic force in NaCl.

42.11 •• When a hypothetical diatomic molecule having atoms 0.8900 nm apart undergoes a rotational transition from the $l = 2$ state to the next lower state, it gives up a photon having energy 8.800×10^{-4} eV. When the molecule undergoes a vibrational transition from one energy state to the next lower energy state, it gives up 0.2590 eV. Find the force constant of this molecule.

Section 42.3 Structure of Solids

42.12 • Potassium bromide (KBr) has a density of $2.75 \times 10^3\ \text{kg/m}^3$ and the same crystal structure as NaCl. The mass of a potassium atom is 6.49×10^{-26} kg, and the mass of a bromine atom is 1.33×10^{-25} kg. (a) Calculate the average spacing between adjacent atoms in a KBr crystal. (b) How does the value calculated in part (a) compare with the spacing in NaCl (see Exercise 42.13)? Is the relationship between the two values qualitatively what you would expect? Explain.

42.13 • **Density of NaCl**. The spacing of adjacent atoms in a crystal of sodium chloride is 0.282 nm. The mass of a sodium atom is 3.82×10^{-26} kg, and the mass of a chlorine atom is 5.89×10^{-26} kg. Calculate the density of sodium chloride.

Section 42.4 Energy Bands

42.14 • The gap between valence and conduction bands in diamond is 5.47 eV. (a) What is the maximum wavelength of a photon that can excite an electron from the top of the valence band into the conduction band? In what region of the electromagnetic spectrum does this photon lie? (b) Explain why pure diamond is transparent and colorless. (c) Most gem diamonds have a yellow color. Explain how impurities in the diamond can cause this color.

42.15 • The maximum wavelength of light that a certain silicon photocell can detect is 1.11 μm. (a) What is the energy gap (in electron volts) between the valence and conduction bands for this photocell? (b) Explain why pure silicon is opaque.

42.16 • The gap between valence and conduction bands in silicon is 1.12 eV. A nickel nucleus in an excited state emits a gamma-ray photon with wavelength 9.31×10^{-4} nm. How many electrons can be excited from the top of the valence band to the bottom of the conduction band by the absorption of this gamma ray?

Section 42.5 Free-Electron Model of Metals

42.17 • Calculate the density of states $g(E)$ for the free-electron model of a metal if $E = 7.8$ eV and $V = 1.0\ \text{cm}^3$. Express your answer in units of states per electron volt.

42.18 • The Fermi energy of sodium is 3.23 eV. (a) Find the average energy E_{av} of the electrons at absolute zero. (b) What is the speed of an electron that has energy E_{av}? (c) At what Kelvin temperature T is kT equal to E_{F}? (This is called the *Fermi temperature* for the metal. It is approximately the temperature at which molecules in a classical ideal gas would have the same kinetic energy as the fastest-moving electron in the metal.)

42.19 • Consider the density of states in the free-electron model for an energy of 1.60 eV. At what energy will the density of states be (a) doubled and (b) halved?

42.20 • For a metal with kT equal to 25% of the Fermi energy, what is the probability that a state will be occupied by an electron if its energy is (a) 50% of the Fermi energy and (b) 50% greater than the Fermi energy? Are your results consistent with Fig. 42.23?

42.21 • CP Silver has a Fermi energy of 5.48 eV. Calculate the electron contribution to the molar heat capacity at constant volume of silver, C_V, at 282 K. Express your result (a) as a multiple of R and (b) as a fraction of the actual value for silver, $C_V = 25.3\ \text{J/mol} \cdot \text{K}$. (c) Is the value of C_V due principally to the electrons? If not, to what is it due? (*Hint:* See Section 18.4.)

42.22 •• At the Fermi temperature T_{F}, $E_{\text{F}} = kT_{\text{F}}$ (see Exercise 42.18). When $T = T_{\text{F}}$, what is the probability that a state with energy $E = 2E_{\text{F}}$ is occupied?

42.23 •• For a solid metal having a Fermi energy of 8.480 eV, what is the probability, at room temperature, that a state having an energy of 8.540 eV is occupied by an electron?

Section 42.6 Semiconductors

42.24 • Pure germanium has a band gap of 0.67 eV. The Fermi energy is in the middle of the gap. (a) For temperatures of 255 K, 290 K, and 355 K, calculate the probability $f(E)$ that a state at the bottom of the conduction band is occupied. (b) For each temperature in part (a), calculate the probability that a state at the top of the valence band is empty.

42.25 • Germanium has a band gap of 0.67 eV. Doping with arsenic adds donor levels in the gap 0.01 eV below the bottom of the conduction band. At a temperature of 300 K, the probability is 4.4×10^{-4} that an electron state is occupied at the bottom of the conduction band. Where is the Fermi level relative to the conduction band in this case?

Section 42.7 Semiconductor Devices

42.26 •• (a) Suppose a piece of very pure germanium is to be used as a light detector by observing, through the absorption of photons, the increase in conductivity resulting from generation of electron–hole pairs. If each pair requires 0.67 eV of energy, what is the maximum wavelength that can be detected? In what portion of the spectrum does it lie? (b) What are the answers to part (a) if the material is silicon, with an energy requirement of 1.12 eV per pair, corresponding to the gap between valence and conduction bands in that element?

42.27 • CP At a temperature of 290 K, a certain p-n junction has a saturation current $I_{\text{S}} = 0.500$ mA. (a) Find the current at this temperature when the voltage is (i) 1.00 mV, (ii) −1.00 mV, (iii) 100 mV, and (iv) −100 mV. (b) Is there a region of applied voltage where the diode obeys Ohm's law?

42.28 • For a certain p-n junction diode, the saturation current at room temperature (20°C) is 0.950 mA. What is the resistance of this diode when the voltage across it is (a) 85.0 mV and (b) −50.0 mV?

42.29 •• (a) A forward-bias voltage of 15.0 mV produces a positive current of 9.25 mA through a p-n junction at 300 K. What does the positive current become if the forward-bias voltage is reduced to 10.0 mV? (b) For reverse-bias voltages of −15.0 mV and −10.0 mV, what is the reverse-bias negative current?

42.30 •• A p-n junction has a saturation current of 6.40 mA. (a) At a temperature of 300 K, what voltage is needed to produce a positive current of 40.0 mA? (b) For a voltage equal to the negative of the value calculated in part (a), what is the negative current?

42.31 • The saturation current for a junction diode is 10 microamps. Calculate the resistance of the diode at forward and reverse potential differences of 0.20 V when $T = 17$°C.

PROBLEMS

42.32 • When a diatomic molecule undergoes a transition from the $l = 2$ to the $l = 1$ rotational state, a photon with wavelength 54.3 μm is emitted. What is the moment of inertia of the molecule for an axis through its center of mass and perpendicular to the line connecting the nuclei?

42.33 •• CP (a) The equilibrium separation of the two nuclei in an NaCl molecule is 0.24 nm. If the molecule is modeled as charges $+e$ and $-e$ separated by 0.24 nm, what is the electric dipole moment of the molecule (see Section 21.7)? (b) The measured electric dipole moment of an NaCl molecule is $3.0 \times 10^{-29}\ \text{C} \cdot \text{m}$. If this dipole moment arises from point charges $+q$ and $-q$ separated by 0.24 nm, what is q? (c) A definition of the *fractional ionic character* of the bond is q/e. If the sodium atom has charge $+e$ and the chlorine atom has charge $-e$, the fractional ionic character would be equal to 1. What is the actual fractional ionic character for the bond in NaCl? (d) The equilibrium distance between nuclei in the hydrogen iodide (HI) molecule is 0.16 nm, and the measured electric dipole moment of the molecule is $1.5 \times 10^{-30}\ \text{C} \cdot \text{m}$. What is the fractional ionic character for the bond in HI? How does your answer compare to that for NaCl calculated in part (c)? Discuss reasons for the difference in these results.

42.34 • When a NaF molecule makes a transition from the $l = 3$ to the $l = 2$ rotational level with no change in vibrational quantum number or electronic state, a photon with wavelength 3.83 mm is emitted. A sodium atom has mass 3.82×10^{-26} kg, and a fluorine atom has mass 3.15×10^{-26} kg. Calculate the equilibrium separation between the nuclei in a NaF molecule. How does your answer compare with the value for NaCl given in Section 42.1? Is this result reasonable? Explain.

42.35 •• CP Consider a gas of diatomic molecules (moment of inertia I) at an absolute temperature T. If E_{g} is a ground-state energy and E_{ex} is the energy of an excited state, then the Maxwell–Boltzmann distribution (see Section 39.4) predicts that the ratio of the numbers of molecules in the two states is $n_{\text{ex}}/n_{\text{g}} = e^{-(E_{\text{ex}} - E_{\text{g}})/kT}$. (a) Explain why the ratio of the number of molecules in the lth rotational energy *level* to the number of molecules in the ground-state ($l = 0$) rotational level is

$$\frac{n_l}{n_0} = (2l + 1)e^{-[l(l+1)\hbar^2]/2IkT}$$

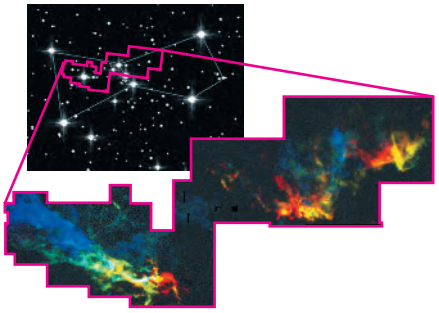
(*Hint:* For each value of l, how many states are there with different values of m_l?) (b) Determine the ratio n_l/n_0 for a gas of CO molecules at 300 K for (i) $l = 1$; (ii) $l = 2$; (iii) $l = 10$; (iv) $l = 20$; (v) $l = 50$. The moment of inertia of the CO molecule is given in Example 42.2 (Section 42.2). (c) Your results in part (b) show that as l is increased, the ratio n_l/n_0 first increases and then decreases. Explain why.

42.36 ••• **CALC** Part (a) of Problem 42.35 gives an equation for the number of diatomic molecules in the lth rotational level to the number in the ground-state rotational level. (a) Derive an expression for the value of l for which this ratio is the largest. (b) For the CO molecule at $T = 300$ K, for what value of l is this ratio a maximum? (The moment of inertia of the CO molecule is given in Example 42.2.)

42.37 •• Semiconductor devices can radiate heat up to a maximum power rating. If the device dissipates more power than that limit, it will heat up and explode, melt, or simply break down. A typical power rating for a diode is 500 mW. The power dissipated by any two-terminal device is the product of the current through it and the voltage across it. (a) Use Eq. (42.22) to estimate the power dissipated by a diode when it is forward biased at $V = 0.6$ V at a temperature of 300 K. The saturation current is 1.2×10^{-11} A. (b) Estimate to one significant figure the forward voltage at which the device will dissipate 500 mW. (c) Then what is the current?

42.38 •• Our galaxy contains numerous *molecular clouds,* regions many light-years in extent in which the density is high enough and the temperature low enough for atoms to form into molecules. Most of the molecules are H_2, but a small fraction of the molecules are carbon monoxide (CO). Such a molecular cloud in the constellation Orion is shown in **Fig. P42.38**. The upper image was made with an ordinary visible-light telescope; the lower image shows the molecular cloud in Orion as imaged with a radio telescope tuned to a wavelength emitted by CO in a rotational transition. The different colors in the radio image indicate regions of the cloud that are moving either toward us (blue) or away from us (red) relative to the motion of the cloud as a whole, as determined by the Doppler shift of the radiation. (Since a molecular cloud has about 10,000 hydrogen molecules for each CO molecule, it might seem more reasonable to tune a radio telescope to emissions from H_2 than to emissions from CO. Unfortunately, it turns out that the H_2 molecules in molecular clouds do not radiate in either the radio or visible portions of the electromagnetic spectrum.) (a) Using the data in Example 42.2 (Section 42.2), calculate the energy and wavelength of the photon emitted by a CO molecule in an $l = 1 \rightarrow l = 0$ rotational transition. (b) As a rule, molecules in a gas at temperature T will be found in a certain excited rotational energy level, provided the energy of that level is no higher than kT (see Problem 42.35). Use this rule to explain why astronomers can detect radiation from CO in molecular clouds even though the typical temperature of a molecular cloud is a very low 20 K.

Figure **P42.38**



42.39 • The force constant for the internuclear force in a hydrogen molecule (H_2) is $k' = 576$ N/m. A hydrogen atom has mass 1.67×10^{-27} kg. Calculate the zero-point vibrational energy for H_2 (that is, the vibrational energy the molecule has in the $n = 0$ ground vibrational level). How does this energy compare in magnitude with the H_2 bond energy of -4.48 eV?

42.40 • When an OH molecule undergoes a transition from the $n = 0$ to the $n = 1$ vibrational level, its internal vibrational energy increases by 0.463 eV. Calculate the frequency of vibration and the force constant for the interatomic force. (The mass of an oxygen atom is 2.66×10^{-26} kg, and the mass of a hydrogen atom is 1.67×10^{-27} kg.)

42.41 • The hydrogen iodide (HI) molecule has equilibrium separation 0.160 nm and vibrational frequency 6.93×10^{13} Hz. The mass of a hydrogen atom is 1.67×10^{-27} kg, and the mass of an iodine atom is 2.11×10^{-25} kg. (a) Calculate the moment of inertia of HI about a perpendicular axis through its center of mass. (b) Calculate the wavelength of the photon emitted in each of the following vibration-rotation transitions: (i) $n = 1, l = 1 \rightarrow n = 0, l = 0$; (ii) $n = 1, l = 2 \rightarrow n = 0, l = 1$; (iii) $n = 2, l = 2 \rightarrow n = 1, l = 3$.

42.42 • Suppose the hydrogen atom in HF (see the Bridging Problem for this chapter) is replaced by an atom of deuterium, an isotope of hydrogen with a mass of 3.34×10^{-27} kg. The force constant is determined by the electron configuration, so it is the same as for the normal HF molecule. (a) What is the vibrational frequency of this molecule? (b) What wavelength of light corresponds to the energy difference between the $n = 1$ and $n = 0$ levels? In what region of the spectrum does this wavelength lie?

42.43 •• Compute the Fermi energy of potassium by making the simple approximation that each atom contributes one free electron. The density of potassium is 851 kg/m^3, and the mass of a single potassium atom is 6.49×10^{-26} kg.

42.44 •• **CALC** The one-dimensional calculation of Example 42.4 (Section 42.3) can be extended to three dimensions. For the three-dimensional fcc NaCl lattice, the result for the potential energy of a pair of Na^+ and Cl^- ions due to the electrostatic interaction with all of the ions in the crystal is $U = -\alpha e^2/4\pi\epsilon_0 r$, where $\alpha = 1.75$ is the *Madelung constant*. Another contribution to the potential energy is a repulsive interaction at small ionic separation r due to overlap of the electron clouds. This contribution can be represented by A/r^8, where A is a positive constant, so the expression for the total potential energy is

$$U_{\text{tot}} = -\frac{\alpha e^2}{4\pi\epsilon_0 r} + \frac{A}{r^8}$$

(a) Let r_0 be the value of the ionic separation r for which U_{tot} is a minimum. Use this definition to find an equation that relates r_0 and A, and use this to write U_{tot} in terms of r_0. For NaCl, $r_0 = 0.281$ nm. Obtain a numerical value (in electron volts) of U_{tot} for NaCl. (b) The quantity $-U_{\text{tot}}$ is the energy required to remove an Na^+ ion and a Cl^- ion from the crystal. Forming a pair of neutral atoms from this pair of ions involves the release of 5.14 eV (the ionization energy of Na) and the expenditure of 3.61 eV (the electron affinity of Cl). Use the result of part (a) to calculate the energy required to remove a pair of neutral Na and Cl atoms from the crystal. The experimental value for this quantity is 6.39 eV; how well does your calculation agree?

42.45 ••• Metallic lithium has a bcc crystal structure. Each unit cell is a cube of side length $a = 0.35$ nm. (a) For a bcc lattice, what is the number of atoms per unit volume? Give your answer in terms of a. (*Hint:* How many atoms are there per unit cell?) (b) Use the result of part (a) to calculate the zero-temperature Fermi energy E_{F0} for metallic lithium. Assume there is one free electron per atom.

42.46 •• **DATA** To determine the equilibrium separation of the atoms in the HCl molecule, you measure the rotational spectrum of HCl. You find that the spectrum contains these wavelengths (among others): 60.4 μm, 69.0 μm, 80.4 μm, 96.4 μm, and 120.4 μm. (a) Use your measured wavelengths to find the moment of inertia of the HCl molecule about an axis through the center of mass and perpendicular to the line joining the two nuclei. (b) The value of l changes by ± 1 in rotational transitions. What value of l for the upper level of the transition gives rise to each of these wavelengths? (c) Use your result of part (a) to calculate the equilibrium separation of the atoms in the HCl molecule. The mass of a chlorine atom is 5.81×10^{-26} kg, and the mass of a hydrogen atom is 1.67×10^{-27} kg. (d) What is the longest-wavelength line in the rotational spectrum of HCl?

42.47 •• **DATA** The table gives the occupation probabilities $f(E)$ as a function of the energy E for a solid conductor at a fixed temperature T.

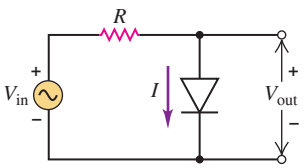
$f(E)$	0.064	0.173	0.390	0.661	0.856	0.950
E (eV)	3.0	2.5	2.0	1.5	1.0	0.5

To determine the Fermi energy of the solid material, you are asked to analyze this information in terms of the Fermi–Dirac distribution. (a) Graph the values in the table as E versus $\ln\{[1/f(E)] - 1\}$. Find the slope and y-intercept of the best-fit straight line for the data points when they are plotted this way. (b) Use your results of part (a) to calculate the temperature T and the Fermi energy of the material.

42.48 •• **DATA** A p-n junction is part of the control mechanism for a wind turbine that is used to generate electricity. The turbine has been malfunctioning, so you are running diagnostics. You can remotely change the bias voltage V applied to the junction and measure the current through the junction. With a forward-bias voltage of $+5.00$ mV, the current is $I_f = 0.407$ mA. With a reverse-bias voltage of -5.00 mV, the current is $I_r = -0.338$ mA. Assume that Eq. (42.22) accurately represents the current–voltage relationship for the junction, and use these two results to calculate the temperature T and saturation current I_S for the junction. [*Hint:* In your analysis, let $x = e^{eV/kT}$. Apply Eq. (42.22) to each measurement and obtain a quadratic equation for x.]

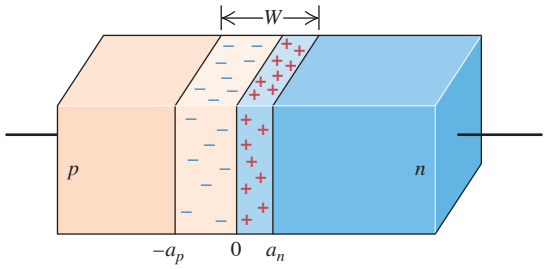
42.49 •• The circuit shown in **Fig. P42.49** is called a half-wave rectifier. The triangular symbol with the flat lower edge represents a diode, such that a forward bias drives the current downward. Use Eq. (42.22) to study the behavior of this circuit. Use a typical value for the saturation current: $I_S = 5 \times 10^{-13}$ A. Use a temperature of 300 K. (a) Any predicted current less than 1 μA can be considered a zero current. Determine the minimum voltage V_{out} for which the current I exceeds 1 μA. (b) The current increases dramatically for diode voltages that exceed that value. At a given diode voltage V_{out}, the diode has an effective resistance given by $R_D = V_{out}/I$. Use Eq. (42.22) to estimate the diode voltage $V_{out} = V_{max}$ for which $R_D = 100\ \Omega$. (c) What is the current I in that case? (d) If the resistance R is greater than 10 kΩ, then $R_D \ll R$ for larger currents, and there is a negligible additional voltage drop across the diode. This effectively pins $V_{out} = V_{max}$ for all input voltages $V_{in} > V_{max}$. With this perspective, estimate how this circuit would respond to a sinusoidal input voltage with amplitude 5 V.

Figure **P42.49**



42.50 ••• **CP** **CALC** A p-n junction includes p-type silicon, with donor atom density N_D, adjacent to n-type silicon, with acceptor atom density N_A. Near the junction, free electrons from the n side have diffused into the p side while free holes from the p side have diffused into the n side, leaving a "depletion region" of width W where there are no free charge carriers (**Fig. P42.50**). The width of the depletion region on the n side is a_n, and the width of the depletion region on the p side is a_p. Each depleted region has a constant charge density with magnitude equal to the corresponding donor atom density. Assume there is no net charge outside the depleted regions. Define an x-axis pointing from the p side toward the n side with the origin at the center of the junction. An electric field $\vec{E} = -E\hat{\imath}$ has developed in the depletion region, stabilizing the diffusion. (a) Use Gauss's law to determine the electric field on the p side of the junction, for $-a_p \le x \le 0$. (*Hint:* Use a cylindrical Gaussian surface parallel to the x-axis, with the left end outside the depletion region (where $\vec{E} = 0$) and the right side inside the region.) Note that the dielectric constant of silicon is $K = 11.7$. (b) Determine the electric field on the n side of the junction, for $0 \le x \le a_n$.

Figure **P42.50**



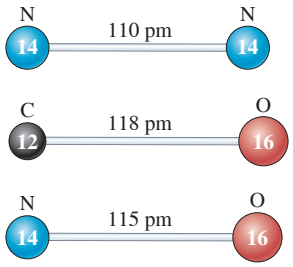
(c) Use continuity at $x = 0$ to determine a relationship between the depths a_p and a_n and in terms of N_A and N_D. (d) The electric potential $V(x)$ may be obtained by integrating the electric field, as shown in Eq. (23.18). Determine $V(x)$ in the region $-a_p \le x \le 0$ using the convention that $V(x) = 0$ for $x < -a_p$. (e) Similarly, determine $V(x)$ in the region $0 \le x \le a_n$. (f) What is the "barrier potential" $V_b = V(a_n) - V(-a_p)$? (g) The p side is doped with boron atoms with density $N_A = 1.00 \times 10^{16}$ cm^{-3}. The n side is doped with arsenic atoms with density $N_D = 5.00 \times 10^{16}$ cm^{-3}. The n side depletion depth is $a_n = 55.0$ nm. What is the p side depletion depth a_p? (h) What is the peak magnitude of the electric field, at $x = 0$? (i) What is the value of the barrier potential V_b?

42.51 ••• **CP** Consider the CO_2 molecule shown in Fig. 42.10c. The oxygen molecules have mass M_O and the carbon atom has mass M_C. Parameterize the positions of the left oxygen atom, the carbon atom, and the right oxygen atom using x_1, x_2, and x_3 as the respective rightward deviations from equilibrium. Treat the bonds as Hooke's-law springs with common spring constant k'. (a) Use Newton's second law to obtain expressions for $M_i\ddot{x}_i = M_i\, d^2x/dt^2$ in each case $i = 1, 2, 3$, where $M_{1,2,3} = (M_O, M_C, M_O)$. (*Note:* We represent time derivatives using dots.) Assume $X_C = 0$ at $t = 0$. (b) To ascertain the motion of the asymmetric stretching mode, set $x_1 = x_3 \equiv X_O$ and set $x_2 \equiv X_C$. Write the two independent equations that remain from your previous result. (c) Eliminate the sum $X_O + X_C$ from your equations. Use what remains to ascertain X_C in terms of X_O. (d) Substitute your expression for X_C into your equation for X_O to derive a harmonic oscillator equation $M_{eff}\ddot{X}_O = -k\ddot{X}_O$. What is M_{eff}? (e) This equation has the solution $X_O(t) = A\cos(\omega t)$. What is the angular frequency ω? (f) Using the experimentally determined spring constant $k' = 1860$ N/m and the atomic masses $M_C = 12$ u and $M_O = 16$ u, where u $= 1.6605 \times 10^{-27}$ kg, to determine the oscillation frequency $f = \omega/2\pi$.

CHALLENGE PROBLEMS

42.52 ••• A colorless, odorless gas in a tank needs to be identified. The gas is nitrogen (N_2), carbon monoxide (CO), or nitric oxide (NO). (See **Fig. P42.52**.) White light is passed through a sample to obtain an absorption spectrum. Among others, the following wavelengths present dark bands on the spectrum: $\lambda = 4.2680\ \mu$m, $4.2753\ \mu$m, $4.2826\ \mu$m, $4.2972\ \mu$m, and $4.3046\ \mu$m. These represent transitions from the lowest rotational states. (a) What photon energies correspond to these wavelengths? Specify your answers to five significant figures. Note that the speed of light is 2.9979×10^8 m/s and $h = 4.1357 \times 10^{-15}$ eV · s. (b) Which one of the absorbed energies corresponds to transitions *from* an $l = 0$ state of the gas? (*Hint:* Use Eq. (42.9) along with the transition rules as a guide.) (c) Which one of the absorbed energies corresponds to a transition *to* an $l = 0$ state? (d) Determine the value of $\hbar\sqrt{k'/m_r}$, where k' is the effective spring constant of the molecule and m_r is its reduced mass. (*Hint:* Use the previous two results and Eq. (42.9).) (e) Determine the molecule's moment of inertia I. (f) The atomic masses and the bond

Figure P42.52



lengths for the three molecule options are shown in Fig. P42.52, where u = 1.6605 × 10^{-27} kg is one atomic mass unit. Based on the data, what is the mystery gas? (g) What is the reduced mass m_r in terms of u? (h) What is the spring constant k'?

42.53 ••• **CALC** Consider a system of N free electrons within a volume V. Even at absolute zero, such a system exerts a pressure p on its surroundings due to the motion of the electrons. To calculate this pressure, imagine that the volume increases by a small amount dV. The electrons will do an amount of work $p\,dV$ on their surroundings, which means that the total energy E_{tot} of the electrons will change by an amount $dE_{tot} = -p\,dV$. Hence $p = -dE_{tot}/dV$. (a) Show that the pressure of the electrons at absolute zero is

$$p = \frac{3^{2/3}\pi^{4/3}\hbar^2}{5m}\left(\frac{N}{V}\right)^{5/3}$$

(b) Evaluate this pressure for copper, which has a free-electron concentration of 8.45 × 10^{28} m^{-3}. Express your result in pascals and in atmospheres. (c) The pressure you found in part (b) is extremely high. Why, then, don't the electrons in a piece of copper simply explode out of the metal?

42.54 ••• **CALC** When the pressure p on a material increases by an amount Δp, the volume of the material will change from V to $V + \Delta V$, where ΔV is negative. The *bulk modulus* B of the material is defined to be the ratio of the pressure change Δp to the absolute value $|\Delta V/V|$ of the fractional volume change. The greater the bulk modulus, the greater the pressure increase required for a given fractional volume change, and the more incompressible the material (see Section 11.4). Since $\Delta V < 0$, the bulk modulus can be written as $B = -\Delta p/(\Delta V/V_0)$. In the limit that the pressure and volume changes are very small, this becomes

$$B = -V\frac{dp}{dV}$$

(a) Use the result of Challenge Problem 42.53 to show that the bulk modulus for a system of N free electrons in a volume V at low temperatures is $B = \frac{5}{3}p$. (*Hint:* The quantity p in the expression $B = -V(dp/dV)$ is the *external* pressure on the system. Can you explain why this is equal to the *internal* pressure of the system itself, as found in Challenge Problem 42.53?) (b) Evaluate the bulk modulus for the electrons in copper, which has a free-electron concentration of 8.45 × 10^{28} m^{-3}. Express your result in pascals. (c) The actual bulk modulus of copper is 1.4 × 10^{11} Pa. Based on your result in part (b), what fraction of this is due to the free electrons in copper? (This result shows that the free electrons in a metal play a major role in making the metal resistant to compression.) What do you think is responsible for the remaining fraction of the bulk modulus?

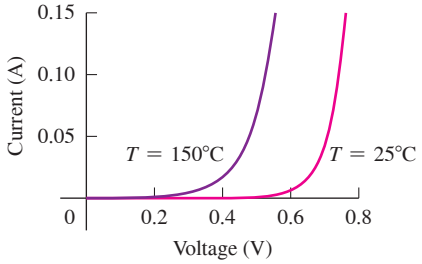
42.55 •• In the discussion of free electrons in Section 42.5, we assumed that we could ignore the effects of relativity. This is not a safe assumption if the Fermi energy is greater than about $\frac{1}{100}mc^2$ (that is, more than about 1% of the rest energy of an electron). (a) Assume that the Fermi energy at absolute zero, as given by Eq. (42.19), is equal to $\frac{1}{100}mc^2$. Show that the electron concentration is

$$\frac{N}{V} = \frac{2^{3/2}m^3c^3}{3000\pi^2\hbar^3}$$

and determine the numerical value of N/V. (b) Is it a good approximation to ignore relativistic effects for electrons in a metal such as copper, for which the electron concentration is 8.45 × 10^{28} m^{-3}? Explain. (c) A *white dwarf star* is what is left behind by a star like the sun after it has ceased to produce energy by nuclear reactions. (Our own sun will become a white dwarf star in another 8 × 10^9 years or so.) A typical white dwarf has mass 2 × 10^{30} kg (comparable to the sun) and radius 6000 km (comparable to that of the earth). The gravitational attraction of different parts of the white dwarf for each other tends to compress the star; what prevents it from compressing is the pressure of free electrons within the star (see Challenge Problem 42.53). Use both of the following assumptions to estimate the electron concentration within a typical white dwarf star: (i) the white dwarf star is made of carbon, which has a mass per atom of 1.99 × 10^{-26} kg; and (ii) all six of the electrons from each carbon atom are able to move freely throughout the star. (d) Is it a good approximation to ignore relativistic effects in the structure of a white dwarf star? Explain.

MCAT-STYLE PASSAGE PROBLEMS

Diode Temperature Sensor. The current–voltage characteristics of a forward-biased *p-n* junction diode depend strongly on temperature, as shown in the figure. As a result, diodes can be used as temperature sensors. In actual operation, the voltage is adjusted to keep the current through the diode constant at a specified value, such as 100 mA, and the temperature is determined from a measurement of the voltage at that current.



42.56 The sensitivity of a diode thermometer depends on how much the voltage changes for a given temperature change, with the current remaining constant. What is the sensitivity for this diode thermometer, operated at 100 mA, for a temperature change from 25°C to 150°C? (a) +0.2 mV/°C; (b) +2.0 mV/°C; (c) −0.2 mV/°C; (d) −2.0 mV/°C.

42.57 Which statement best explains the temperature dependence of the current–voltage characteristics that the graph shows? At higher temperatures: (a) The band gap is larger, so the electron–hole pairs have more energy, which causes the current at a given voltage to be larger. (b) More electrons can move to the conduction band, which causes the current at a given voltage to be larger. (c) All of the electrons in the valence band move to the conduction band, and the diode behaves like a metal and follows Ohm's law. (d) The acceptor and donor impurity atoms are free to move through the material, which causes the current at a given voltage to be larger.

42.58 If the voltage rather than the current is kept constant, what happens as the temperature increases from 25°C to 150°C? (a) At first the current increases, then it decreases. (b) The current increases. (c) The current decreases, eventually approaching zero. (d) The current does not change unless the voltage also changes.

ANSWERS

Chapter Opening Question ?

(i) Venus must radiate energy into space at the same rate that it receives energy in the form of sunlight. However, carbon dioxide (CO_2) molecules in the atmosphere absorb infrared radiation emitted by the surface of Venus and re-emit it toward the ground. This involves a transition between vibrational states of the CO_2 molecule (see Section 42.2). To compensate for this and to maintain the balance between emitted and received energy, the surface temperature of Venus and hence the rate of blackbody radiation from the surface increase.

Key Example √ARIATION Problems

VP42.3.1 (a) 1.240×10^{-26} kg (b) 1.651×10^{-46} kg · m^2
(c) 0, 0.4203 meV, 1.261 meV
VP42.3.2 (a) 2.64×10^{-47} kg · m^2 (b) 1.627×10^{-27} kg (c) 0.127 nm
VP42.3.3 (a) 3.853×10^{-46} kg · m^2 (b) 1.690×10^{-26} kg
(c) 0.1510 nm
VP42.3.4 (a) $f = 6.538 \times 10^{13}$ Hz, $\lambda = 4.585\,\mu$m
(b) $f = 6.527 \times 10^{13}$ Hz, $\lambda = 4.593\,\mu$m
VP42.7.1 (a) $E_F + 0.71kT$ (b) $E_F - 2.2kT$
VP42.7.2 (a) 0.729 (b) 0.170 (c) 0.0187
VP42.7.3 (a) 8.8×10^{-19} J $= 5.5$ eV (b) 1.4×10^6 m/s
(c) 9.9×10^{40} states/J $= 1.6 \times 10^{22}$ states/eV
VP42.7.4 (a) 5.4 eV (b) 5.7 eV
VP42.9.1 (a) 3.79×10^{-5} (b) 5.46×10^{-14}
VP42.9.2 (a) 1.65 (b) 2.72
VP42.9.3 (a) 0.329 eV (b) 0.351 eV
VP42.9.4 (a) 0.0405 (b) 0.0643

Bridging Problem

(a) 0.278 eV
(b) 1.74×10^{-25}